

BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

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Abstract

Large language models memorize training data, creating legal, privacy, and safety risks that demand targeted knowledge removal without full retraining. Current unlearning methods pair unbounded forget objectives—which destroy model coherence—with fixed-weight loss balancing that cannot track the shifting difficulty of retain preservation during training.

We propose BLADE, a bilevel optimization framework for LLM unlearning. The forget objective is *clamped entropy*: per-token output entropy is maximized only up to a fraction τ of maximum entropy $H_{\max} = \log V$, after which the gradient is exactly zero. This gives a bounded loss whose active set shrinks as tokens are forgotten, naturally decelerating optimization. Retain preservation is enforced as an inequality constraint $\mathcal{L}_{\text{ret}} \leq \varepsilon$ via an augmented Lagrangian with an asymmetric dual update—violations ratchet λ up fast, satisfaction lets it decay slowly—nested inside a bilevel structure where the inner loop runs retain repair and the outer loop drives forgetting. All updates are confined to LoRA adapters.

Experiments cover TOFU (Llama-3.2-1B/3B, three forget splits, 5 seeds), MUSE Books and News (Llama-2-7B), and KnowUndo (Llama-2-7B-chat, copyright/privacy). BLADE achieves HM=0.808 on TOFU 1B forget01 versus 0.690 for the strongest baseline, wins all 6 TOFU settings under an LLM judge, and reaches HM=0.823 on MUSE Books—surpassing even the gold retrained model. An ablation confirms that removing any single component causes failure, and dynamics analysis shows a consistent three-phase convergence pattern absent from baselines.

1 Introduction

Large language models memorize substantial portions of their training data (Carlini et al., 2021), raising concerns around copyright, privacy, and retention of hazardous knowledge. *Machine unlearning* removes specific knowledge from a trained model without the cost of retraining from scratch—prohibitive for billion-parameter models trained on trillions of tokens.

The core difficulty is the *forget-retain trade-off*. Aggressive removal of memorized content perturbs representations shared with benign knowledge, degrading the model’s general capabilities. Existing methods handle this through loss balancing: a forget term plus a retain term with fixed weights (Liu et al., 2025; Zhang et al., 2024). Fixed coefficients cannot adapt as training progresses and the difficulty of retaining shifts.

Three failure modes recur in current approaches:

1. **Unbounded forgetting.** Gradient ascent (Jang et al., 2023) and logit-margin losses produce arbitrarily large gradients on highly memorized tokens, causing utility collapse even with retain regularization.
2. **Static tradeoff.** Fixed-weight combinations (GradDiff, NPO) cannot react when retain performance degrades mid-training.
3. **Over-forgetting.** Unclamped entropy maximization (Yuan et al., 2025) and gradient ascent continue to push already-forgotten tokens, wasting gradient budget and destabilizing the model.

BLADE addresses all three through two contributions:

1. **Clamped entropy loss** (Section 3.4): per-token output entropy is maximized up to

a vocabulary-invariant threshold $\tau \cdot H_{\max}$, where $H_{\max} = \log V$. Once a token’s entropy crosses this threshold, its gradient contribution is exactly zero—it has been forgotten and drops out of optimization. The loss is bounded in $[0, \tau \cdot H_{\max}]$ and shrinks monotonically as training succeeds. No reference model is needed.

2. Inequality-constrained bilevel formulation (Section 3.3, Section 3.5): the inner loop runs fast SGD on the retain set, repairing damage before each outer step drives forgetting. Retain performance is enforced as a hard constraint $\mathcal{L}_{\text{ret}} \leq \varepsilon$ via an augmented Lagrangian whose asymmetric dual update ratchets protection up on violation but decays slowly when satisfied. This “repair first, forget carefully” design inverts prior bilevel methods.

3. LoRA parameterization (Section 3.2): all updates are confined to low-rank adapters, limiting how far the unlearning update can push the model from its pre-trained state.

We evaluate on four benchmarks: TOFU (Maini et al., 2024) with Llama-3.2-1B and 3B across three forget splits (5 seeds each), MUSE Books and News (Shi et al., 2025) with Llama-2-7B, and KnowUndo (Tian et al., 2024) with Llama-2-7B-chat on copyright and privacy domains. On TOFU 1B forget01, BLADE achieves HM=0.808 versus the strongest baseline PDU at 0.690 (+0.118). BLADE wins 4 of 6 TOFU settings on automated metrics and all 6 under an LLM judge, which reveals near-zero forget leakage (FL=0.015/2.0) where the best baseline still leaks substantially (FL=0.855). On MUSE Books, BLADE reaches HM=0.823, surpassing both the retrained gold model (0.739) and the best baseline (0.753). On KnowUndo, it achieves the best harmonic mean on both copyright (0.482) and privacy (0.602). On MUSE News—a harder setting with 889 independent articles—BLADE is competitive (0.542) but PDU leads (0.577), pointing to a limitation of global λ on heterogeneous forget sets.

Beyond end-point metrics, we provide a full ablation and dynamics analysis. Training consistently follows a three-phase pattern: warm-up, spike-and-ratchet, smooth conver-

gence. Baselines show recurring retain oscillations with no self-correction. The ablation confirms that each component is necessary: removing any one causes catastrophic failure or significant degradation.

2 Related Work

2.1 Machine Unlearning for LLMs

Machine unlearning was formalized by Cao and Yang (2015) for efficient data removal without full retraining. Bourtole et al. (2021) introduced SISA training, partitioning data to make unlearning tractable, but these approaches target classification and do not transfer to autoregressive language models.

For LLMs, gradient ascent (GA) is the most direct approach: maximize the loss on the forget set (Jang et al., 2023). The loss is unbounded, grows without limit, and rapidly destroys coherence. GradDiff (Yao et al., 2024) combines GA with cross-entropy on retain data, but retain performance still degrades because the forget gradients push weights in destructive directions.

A preference-optimization line began with NPO (Zhang et al., 2024), adapting DPO (Rafailov et al., 2023) by treating forget samples as negative preferences. NPO collapses more slowly than GA but requires a reference model and saturates without converging to a well-defined target. SimNPO (Fan et al., 2025) drops the reference model while remaining competitive. NGDiff (Jin et al., 2025) normalizes the gradient difference between forget and retain so that neither dominates.

At the representation level, RMU (Li et al., 2024) steers intermediate activations toward random targets for forget data while anchoring retain representations. It is effective in its benchmark but lacks the optimization-theoretic grounding of loss-based methods.

What these approaches share is the absence of a *bounded* forget objective with a per-token stopping criterion. GA diverges. NPO damps asymptotically but never reaches zero. Unclamped entropy maximization (Yuan et al., 2025) pushes all tokens toward uniform indefinitely. The inverted hinge of LoKU (Cha et al., 2025) is bounded but operates on token margins, not distributional entropy, and offers no tunable threshold. Our clamped entropy

loss defines a vocabulary-invariant sufficiency threshold and produces exactly zero gradient once a token crosses it.

2.2 Unlearning Benchmarks

Maini et al. (2024) introduced TOFU, built around 200 synthetic author biographies with QA pairs and a gold retrained model. They evaluate via truth ratios, ROUGE, and membership inference. Shi et al. (2025) proposed MUSE, defining six properties (no verbatim/knowledge memorization, no privacy leakage, utility preservation, scalability, sustainability) over two corpora (news articles and Harry Potter) with Llama-2-7b. KnowUndo (Tian et al., 2024) targets copyright and privacy unlearning from LoRA-fine-tuned models.

We evaluate on TOFU, MUSE, and KnowUndo because they provide complementary views: TOFU tests factual associations in a controlled setting, MUSE tests naturalistic memorization at scale with sustainability/scalability stress tests, and KnowUndo tests domain-specific unlearning from adapter-tuned models.

2.3 Parameter-Efficient Unlearning

LoRA (Hu et al., 2022) factorizes weight updates into low-rank matrices, reducing trainable parameters by orders of magnitude. Several works apply LoRA to unlearning: LoKU (Cha et al., 2025) introduces Fisher-informed LoRA initialization with an inverted hinge loss, and WAGLE (Jia et al., 2024a) assigns importance scores for strategic weight masking.

In BLADE, LoRA serves as a *structural constraint*: the low-rank bottleneck limits how much the unlearning update can change the model, preventing catastrophic drift that occurs with full-parameter methods. The inner loop protects retain within this small subspace, while the outer loop drives forgetting through the same bottleneck.

2.4 Bilevel and Constrained Optimization

Bilevel optimization has a long history in operations research (Bertsekas, 1999) and is increasingly used in machine learning via MAML (Finn et al., 2017) and hyperparameter optimization (Franceschi et al., 2018). The augmented Lagrangian method (ALM) (Bertsekas,

1999; Nocedal and Wright, 2006) maintains dual variables for constrained optimization and can achieve exact constraint satisfaction with finite parameters through adaptive dual updates.

Three recent works apply constrained or bilevel optimization to unlearning. BLUR (Reisizadeh et al., 2026) uses retain as the lower-level objective and forget as the upper-level, optimizing through implicit differentiation. PDU (Entesari et al., 2025) uses a linear Lagrangian (no quadratic penalty) with a logit-margin forget loss and symmetric dual updates. Cheng et al. (2026) use an augmented Lagrangian with an equality constraint, a flat single-level optimizer (not bilevel), and evaluation only on vision models.

BLADE differs structurally: (a) we enforce an *inequality* constraint—the model is never penalized for retain loss below ϵ , unlike equality formulations; (b) the dual update is *asymmetric* (fast increase on violation, $10\times$ slower decay), producing one-shot ratchet behavior rather than recurring oscillations; (c) retain is in the *inner loop*, creating a repair-first dynamic that inverts SCRUB and BLUR; and (d) the forget objective is clamped entropy—bounded with a hard per-token deadzone—unlike PDU’s unbounded logit margin or the gradient ascent in BLUR. Jia et al. (2024b) explored second-order unlearning through influence functions (Koh and Liang, 2017), but without bilevel structure or constraint enforcement.

3 Method

We formulate LLM unlearning as constrained bilevel optimization. The inner loop preserves retain performance; the outer loop drives forgetting subject to an augmented Lagrangian constraint. All updates are confined to LoRA adapters attached to a frozen base model.

3.1 Problem Setup

Let M be a pretrained LLM with parameters θ_0 , $\mathcal{D}_{\text{forget}}$ the data to unlearn, and $\mathcal{D}_{\text{retain}}$ data the model should preserve. We seek parameters θ such that (i) the model’s output distribution on $\mathcal{D}_{\text{forget}}$ is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on $\mathcal{D}_{\text{retain}}$ remains below a threshold ϵ .

3.2 LoRA Parameterization

Rather than modifying full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ($q/k/v/o_proj$, $gate/up/down_proj$):

$$W' = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times d} \quad (1)$$

where $r \ll d$ is the rank and α/r is a scaling factor. With $r = 16$ on a 7B model, only $\sim 0.08\%$ of parameters are trainable. This serves two purposes for unlearning: (a) the low-rank subspace limits the expressivity of the update, preventing catastrophic weight drift, and (b) disabling adapters recovers the exact original model, simplifying debugging and rollback.

3.3 Bilevel Formulation

We decompose the problem into two nested optimization levels: *retain in the inner loop, forget in the outer loop*.

Inner loop (retain preservation). For each outer step, run K SGD steps on the retain set:

$$\theta \leftarrow \theta - \eta_{in} \nabla_{\theta} \mathcal{L}_{CE}(M_{\theta}; \mathcal{B}_{ret}) \quad (2)$$

where $\mathcal{B}_{ret} \sim \mathcal{D}_{retain}$ is a mini-batch and $\mathcal{L}_{CE}$ is cross-entropy. The inner loop uses a fast learning rate ($\eta_{in} = 2e-4$) to restore any retain damage from the previous outer step.

Outer loop (constrained forgetting). The outer objective combines the forget loss with an inequality constraint on retain performance, enforced via an augmented Lagrangian:

$$\mathcal{L}_{outer} = \mathcal{L}_{fgt} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2, \quad r = L_{ret} - \varepsilon \quad (3)$$

where $L_{ret} = \mathcal{L}_{CE}(M_{\theta}; \mathcal{B}'_{ret})$ is evaluated on a *fresh* retain batch (separate from the inner loop), λ is the Lagrange multiplier, ρ is the quadratic penalty coefficient, and ε is the retain budget. The outer loop uses a slow learning rate ($\eta_{out} = 3e-5$), keeping each forgetting step small enough for the inner loop to repair.

Repair-first principle. This assignment inverts prior bilevel unlearning methods: SCRUB (Kurmanji et al., 2023) and BLUR (Reisizadeh et al., 2026) put forgetting in the inner loop and retain repair in the outer loop. Our reasoning is different.

Forgetting is inherently destructive. Gradient ascent, entropy maximization, and

preference-based losses all push the model *away* from learned representations, and such updates cause catastrophic collapse when applied without constraint (Zhang et al., 2024). The problem is compounded by the fact that forget and retain knowledge share entangled representations in transformer networks (Cheng et al., 2026)—perturbing forget tokens inevitably disturbs retain performance.

Given this, retain repair belongs in the inner loop as a stabilizing anchor. It runs first and fast ($\eta_{in}/\eta_{out} \approx 7-10\times$), so the model enters each outer step from a state with good retain performance. The outer loop then applies one small forgetting step—sized so the next inner loop can fully repair any collateral damage. This prevents the accumulation of representational damage that makes repair increasingly difficult when the loops are reversed.

3.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations apply nonzero gradients to every token at every step, even those whose distribution is already near-uniform. This produces unnecessarily large perturbations that destabilize bilevel optimization. We introduce *clamped entropy*, which defines when a token is “sufficiently forgotten” and stops optimizing it:

$$\mathcal{L}_{forget} = \frac{1}{T} \sum_{t=1}^T \max(0, \tau \cdot H_{max} - H(p_t)) \quad (4)$$

where $p_t = \text{softmax}(z_t)$ is the output distribution at position t , $H(p_t) = -\sum_v p_t(v) \log p_t(v)$ is its Shannon entropy, $H_{max} = \log V$ is the entropy of a uniform distribution over vocabulary V , and $\tau \in (0, 1)$ is the target fraction.

Vocabulary-invariant threshold. Normalizing by $H_{max} = \log V$ makes τ interpretable across models: $\tau = 0.7$ means “70% of maximum possible uncertainty,” whether the vocabulary has 32K tokens (Llama-2) or 128K tokens (Llama-3). Without this, a fixed entropy threshold would need re-tuning for each tokenizer.

Hard deadzone. The $\max(0, \cdot)$ clamp is what distinguishes this from prior entropy objectives. Once $H(p_t) \geq \tau \cdot H_{max}$, the gradient contribution of token t is *exactly zero*—not

approximately small, but identically zero. Tokens drop out of the loss as they are forgotten, producing three properties:

- **Bounded loss:** $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\text{max}}]$. The outer perturbation magnitude is bounded by construction (Proposition 1), so the inner loop’s K steps can reliably repair retain damage regardless of training stage.
- **Self-stabilizing:** the number of active tokens (those with $H(p_t) < \tau \cdot H_{\text{max}}$) shrinks monotonically during training. The effective gradient norm decreases automatically, decelerating the outer loop without scheduling.
- **Reference-free:** requires only a single forward pass, unlike DPO-style losses (NPO, SimNPO) that need a frozen reference model in memory.

By contrast, unclamped entropy maximization (Yuan et al., 2025) applies gradient to *all* tokens at *all* steps—including those already at 99% of H_{max} —wasting budget and creating unnecessary perturbations to shared representations. Gradient ascent (Jang et al., 2023) is worse still: its loss is unbounded ($-\log p \rightarrow \infty$ as $p \rightarrow 0$), so a single highly-memorized token can dominate the gradient (Proposition 1c).

3.5 Augmented Lagrangian with Asymmetric Dual Update

The multiplier λ controls how much weight retain preservation receives in the outer objective. Rather than treating it as a fixed hyperparameter, we learn it via a dual update rule.

Standard ALM for equality constraints uses symmetric updates and a penalty that is always active (Bertsekas, 1999). We enforce an *inequality* constraint $L_{\text{ret}} \leq \varepsilon$: the quadratic penalty $\frac{\rho}{2}[\max(0, r)]^2$ activates only when violated (Eq. 3). The model is never penalized for achieving retain loss below ε —it is free to improve on retain without opposition. On top of this, we use an *asymmetric* dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho \cdot r & \text{if } r > 0 \text{ (violation)} \\ \lambda + \alpha \rho \cdot r & \text{if } r \leq 0 \text{ (satisfaction)} \end{cases} \quad (5)$$

with $\alpha = 0.1$. The $10\times$ slower decay means that once a retain violation drives λ up, it stays elevated. Without this asymmetry, λ drops back between spikes, causing repeated forget–retain oscillation cycles (as we observe empirically with PDU).

Auto-epsilon. Rather than hand-tuning an absolute ε , we set it as a fraction of the model’s initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)} \quad (6)$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ is the retain loss at inner step k . This adapts ε automatically to the model and dataset.

Convergence criterion. Training terminates when λ plateaus: $|\lambda_t - \lambda_{t-w}| < \delta$ over a window of w outer steps. At this point the constraint is satisfied ($r \leq 0$) and the forget loss contributes negligible gradient (most tokens have entered the deadzone). This gives a stopping rule without held-out validation: a rising λ signals ongoing conflict, while a flat λ confirms equilibrium. In practice, λ plateaus 10–15 steps after the initial ratchet, coinciding with $\mathcal{L}_{\text{fgt}}$ dropping below 1.0.

3.6 Algorithm Summary

4 Experiments

4.1 Setup

Benchmarks. We evaluate on **TOFU** (Maini et al., 2024) (fictitious QA, Llama-3.2-1B/3B, 3 forget splits), **MUSE Books/News** (Shi et al., 2025) (Harry Potter / BBC articles, Llama-2-7B), and **KnowUndo** (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). All results are 5-fold (seeds 42/123/456/789/1024) unless noted.

Composite scores. We report harmonic means that penalize imbalance across axes:

- TOFU: HM = hmean(MU, 1–Prob, 1–ROUGE)
- MUSE: HM = hmean(1–fk, 1–vm, rk)
- KnowUndo: HM = hmean(1–fgt_R, ret_R, MMLU)
- LLM Judge: HM_J = hmean(1–FL/2, RA/2, rRQ/2)

LLM judge. We use Claude Opus 4.7 to score generations on a 0–2 scale: Forget Leakage (FL↓), Retain Accuracy (RA↑), and retain Response Quality (rRQ↑). Token-level metrics like ROUGE miss subtle leakage that a judge can detect from semantic content.

Algorithm 1 BLADE: Bilevel Augmented Lagrangian Unlearning

Require: Model M with frozen weights θ_0 , $\mathcal{D}_{\text{forget}}$, $\mathcal{D}_{\text{retain}}$

Require: Hyperparams: K , η_{in} , η_{out} , ε_{mul} , ρ , λ_0 , α , τ , T

```
1: Attach LoRA adapters to  $M$ ;  $\theta \leftarrow$  LoRA  
   params  
2:  $\lambda \leftarrow \lambda_0$   
3:  $\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)}(\mathcal{D}_{\text{retain}})$   
4: for  $t = 1$  to  $T$  do  
5:   Inner loop (retain repair):  
6:   for  $k = 1$  to  $K$  do  
7:      $\theta \leftarrow \theta - \eta_{\text{in}} \nabla_{\theta} \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}_{\text{ret}})$   
8:   end for  
9:   Forget loss (clamped entropy):  
10:   $\mathcal{L}_{\text{fgt}} \leftarrow \frac{1}{T} \sum_t \max(0, \tau \log V - H(p_t))$   
11:  Outer loss (augmented Lagrangian):  
12:   $r \leftarrow \mathcal{L}_{\text{CE}}(M_{\theta}; \mathcal{B}'_{\text{ret}}) - \varepsilon$   
13:   $\mathcal{L}_{\text{outer}} \leftarrow \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2$   
14:   $\theta \leftarrow \theta - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\theta} \mathcal{L}_{\text{outer}})$   
15:  Dual update (asymmetric ratchet):  
16:  if  $r > 0$  then  
17:     $\lambda \leftarrow \lambda + \rho \cdot r$   
18:  else  
19:     $\lambda \leftarrow \lambda + \alpha \rho \cdot r$   
20:  end if  
21: end for  
22: return  $M$  with trained LoRA adapters
```

4.2 TOFU (Llama-3.2-1B-Instruct)

4.3 TOFU (Llama-3.2-3B-Instruct)

4.4 MUSE Books (Llama-2-7B)

4.5 MUSE News (Llama-2-7B)

4.6 KnowUndo (Llama-2-7B-chat)

All methods 5-fold except BLURNPO (OOM). MMLU evaluated once at seed=42, used as constant for HM computation.

4.7 Sustainability and Scalability (MUSE News)

Scalability: single-shot unlearning on progressively larger forget sets (1–4× the 889-sample standard). Sustainability: sequential unlearning where each step applies the method on a new 889-sample forget set starting from the previous checkpoint.

PDU collapses at 4× scale (HM=0.005, retain destroyed) and after 2 sequential steps

(HM 0.58→0.02). BLADE degrades gracefully: scalability HM stays 0.542–0.552 through 3×, sustainability HM holds 0.542–0.525 through 4 sequential steps. The bounded loss and LoRA bottleneck prevent the accumulated representational damage that destroys PDU under repeated application.

4.8 Ablation Study

The ALM is the critical component: without it, the model collapses at step 31 (HM=0.233). LoRA provides essential implicit regularization—full fine-tuning achieves aggressive forgetting but catastrophic retain collapse (ret 0.658→0.306). Among forget losses, only clamped entropy works within the bilevel framework; GA, NPO, and logit margin all fail catastrophically even with full ALM protection. The inner loop adds a safety margin (+0.027 retain) but is not strictly required—the ALM constraint alone provides strong protection. Swapping the bilevel structure (forgetting in the inner loop) causes λ to diverge because the auto-epsilon threshold exceeds what clamped entropy can produce.

4.9 Training Dynamics

Across all settings (Figure 1), BLADE follows a three-phase pattern: (1) **Warm-up**: forget loss declines steadily while retain stays near ε . (2) **Spike and ratchet**: accumulated forgetting disrupts shared representations, spiking retain loss; the asymmetric update ratchets λ from ~ 1 to 2.2–4.5, permanently elevating protection. The final λ scales with task difficulty without hyperparameter change: 2.4 (fgt01) → 3.6 (fgt05) → 4.5 (fgt10). (3) **Smooth convergence**: the forget loss approaches zero with no further retain violations.

PDU (Figure 2) shows qualitatively different behavior: its unbounded logit-margin loss starts at 700+ and produces recurring retain spikes throughout training. The symmetric dual update offers no lasting protection— λ drops as fast as it rises, producing oscillatory cycles that never resolve. These oscillations map directly to worse final metrics.

5 Conclusion

We presented BLADE, a bilevel framework for LLM unlearning built on two ideas. First,

Table 1: TOFU 1B full results (5 seeds). HM = hmean(MU, 1-Prob, 1-ROUGE). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.597	.166	.414	.655	.599	.127	.383	.676	.591	.116	.379	.677
GradAscent	.596±.00	.477±.01	.456±.01	.553±.01	.008±.01	.003±.01	.112±.08	.022±.04	.000±.00	.000±.00	.001±.00	.000±.00
GradDiff	.581±.01	.421±.04	.464±.02	.564±.01	.453±.00	.073±.00	.375±.01	.614±.01	.435±.00	.047±.00	.339±.01	.617±.00
NPO	.596±.00	.474±.01	.438±.02	.560±.01	.454±.01	.245±.01	.308±.01	.603±.01	.393±.03	.213±.01	.210±.01	.590±.02
SimNPO	.594±.00	.858±.01	.734±.02	.240±.01	.596±.00	.848±.00	.741±.01	.248±.00	.597±.00	.842±.00	.734±.01	.255±.00
RMU	.557±.00	.414±.01	.415±.00	.576±.00	.546±.00	.369±.01	.423±.01	.583±.00	.572±.00	.106±.02	.322±.01	.691±.01
BLURNPO	.598±.00	.676±.01	.588±.02	.417±.01	.518±.02	.475±.07	.407±.05	.541±.04	.089±.14	.087±.04	.253±.09	.154±.20
PDU	.602±.00	.186±.01	.313±.01	.690±.01	.588±.00	.073±.02	.214±.03	.740±.01	.592±.00	.004±.00	.065±.01	.797±.00
BLADE	.599±.00	.002±.00	.038±.00	.808±.00	.597±.00	.015±.00	.054±.01	.800±.00	.593±.01	.009±.00	.039±.01	.803±.00

Table 2: LLM Judge on TOFU 1B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	FL↓	Forget 1%			FL↓	Forget 5%			FL↓	Forget 10%		
		RA↑	rRQ↑	HM↑		RA↑	rRQ↑	HM↑		RA↑	rRQ↑	HM↑
GradAscent	1.11±.05	1.66±.01	1.99±.00	.674±.02	0.07±.07	0.10±.09	0.51±.43	.117±.10	0.00±.00	0.00±.00	0.00±.00	.000±.00
GradDiff	1.21±.07	1.54±.04	1.97±.01	.620±.02	0.65±.03	1.01±.01	1.68±.01	.643±.00	0.46±.01	0.91±.02	1.53±.04	.625±.01
NPO	1.05±.08	1.67±.01	1.99±.00	.696±.03	0.44±.02	1.14±.05	1.89±.02	.731±.01	0.44±.02	0.94±.07	1.82±.04	.665±.03
SimNPO	1.76±.03	1.66±.01	1.99±.00	.288±.03	1.58±.01	1.69±.01	1.99±.00	.435±.01	1.58±.02	1.72±.02	1.99±.00	.435±.02
RMU	0.81±.05	1.39±.01	1.93±.01	.721±.01	0.72±.04	1.36±.02	1.91±.01	.736±.01	0.29±.04	1.50±.01	1.98±.00	.854±.01
BLURNPO	1.40±.05	1.67±.01	1.99±.00	.541±.03	0.86±.08	1.37±.06	1.98±.01	.709±.01	0.26±.24	0.36±.38	0.73±.64	.235±.21
PDU	0.86±.06	1.60±.01	1.97±.00	.746±.02	0.37±.08	1.56±.02	1.96±.01	.850±.01	0.04±.01	1.57±.01	1.98±.01	.906±.00
BLADE	0.02±.02	1.65±.01	1.98±.01	.929±.01	0.07±.02	1.64±.02	1.98±.01	.919±.00	0.04±.02	1.63±.02	1.98±.00	.919±.01

clamped entropy: a forget loss with a hard per-token deadzone at $\tau \cdot \log V$, giving bounded and self-stabilizing forgetting where forgotten tokens produce exactly zero gradient. Second, an inequality-constrained bilevel formulation where retain repair runs in the inner loop and forgetting in the outer—the inverse of prior bilevel unlearning—with an augmented Lagrangian whose asymmetric dual update permanently ratchets protection from a single retain spike. All updates stay within LoRA adapters.

On TOFU across two model scales and three forget splits (5 seeds), BLADE outperforms baselines in the majority of settings, with the largest gains on small forget sets where the tradeoff is hardest (forget01 1B: +0.118 HM over PDU). An LLM judge confirms near-zero forget leakage where baselines retain substantial residual knowledge. On MUSE Books, BLADE achieves HM=0.818 (5-fold), the highest among all tested methods. On KnowUndo, it achieves the best harmonic mean on both copyright and privacy. On MUSE News, BLADE is competitive but does not dominate, showing that distributed multi-document settings remain challenging for methods with a single global constraint.

The ablation confirms that each component is load-bearing: removing the ALM collapses the model, removing LoRA causes catastrophic drift, and replacing clamped entropy

with any unbounded loss fails even with full bilevel protection. Training dynamics reveal a three-phase convergence pattern—warm-up, spike-and-ratchet, smooth convergence—that is consistent across all scales and datasets, while baselines exhibit uncontrolled retain oscillations.

Limitations. BLADE has more hyperparameters than single-loss baselines (K , ρ , τ , ϵ_{mul}), though auto-epsilon and the observed robustness to ρ and K reduce the practical tuning surface. On MUSE News, a single global λ is insufficient when forget difficulty varies across documents. The bilevel structure adds cost ($K=3$ inner steps per outer step), roughly 4× the wall time of single-loop methods.

Future work. Per-document or per-cluster constraint enforcement could address the MUSE News limitation. Evaluation on safety-critical settings and larger instruction-tuned models (>8B) is a natural next step. The three-phase dynamics suggest connections to phase transitions in learning that warrant further investigation.

Table 3: TOFU 3B full results (5 seeds). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.663	.179	.409	.656	.659	.130	.388	.698	.661	.115	.382	.698
GradAscent	.668±.00	.564±.01	.525±.01	.509±.01	.481±.02	.140±.02	.333±.01	.632±.01	.000±.00	.000±.00	.003±.00	.000±.00
GradDiff	.659±.00	.569±.02	.578±.03	.483±.02	.566±.01	.147±.01	.395±.01	.653±.00	.532±.02	.079±.01	.361±.02	.662±.01
NPO	.668±.00	.553±.01	.494±.01	.525±.01	.542±.01	.219±.00	.359±.01	.640±.00	.549±.01	.237±.01	.353±.06	.640±.02
SimNPO	.656±.00	.920±.01	.830±.02	.151±.01	.657±.00	.901±.00	.818±.01	.175±.00	.652±.00	.889±.00	.808±.00	.191±.00
RMU	.657±.00	.861±.00	.715±.01	.246±.00	.644±.00	.602±.00	.534±.00	.483±.00	.647±.00	.390±.01	.472±.00	.591±.00
BLURNPO	.667±.00	.822±.02	.788±.02	.253±.02	.640±.02	.683±.08	.593±.09	.412±.08	.385±.17	.245±.16	.314±.09	.502±.10
PDU	.692±.00	.168±.01	.283±.01	.742±.00	.687±.00	.009±.00	.085±.01	.843±.00	.680±.01	.000±.00	.029±.01	.857 ±.00
BLADE	.654±.00	.000 ±.00	.015 ±.01	.846 ±.00	.655±.00	.007 ±.01	.028 ±.02	.842±.01	.647±.00	.005±.00	.038 ±.01	.836±.01

Table 4: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	1.34±.09	1.85±.00	2.00±.00	.587±.05	0.69±.04	1.45±.02	1.98±.01	.766±.01	0.00±.00	0.00±.00	0.00±.00	.000±.00
GradDiff	1.50±.07	1.82±.02	2.00±.00	.490±.04	0.83±.01	1.29±.02	1.71±.06	.677±.01	0.66±.03	1.16±.06	1.43±.20	.648±.04
NPO	1.23±.04	1.84±.00	2.00±.00	.640±.02	0.74±.04	1.55±.01	2.00±.00	.774±.01	0.61±.08	1.52±.04	1.99±.01	.796±.01
SimNPO	1.83±.05	1.84±.01	2.00±.00	.220±.06	1.74±.01	1.85±.01	2.00±.00	.305±.01	1.71±.01	1.86±.01	2.00±.00	.331±.01
RMU	1.59±.04	1.81±.00	2.00±.00	.432±.03	1.11±.03	1.70±.01	1.99±.00	.679±.01	0.85±.02	1.70±.01	1.99±.00	.765±.01
BLURNPO	1.73±.12	1.85±.00	2.00±.00	.308±.11	1.40±.15	1.71±.05	2.00±.00	.538±.08	0.73±.35	1.05±.51	1.51±.60	.550±.16
PDU	0.62±.04	1.72±.01	1.97±.00	.828±.01	0.07±.01	1.75±.01	1.98±.00	.941±.00	0.01 ±.01	1.73±.03	1.95±.04	.940±.01
BLADE	0.01 ±.01	1.85±.00	2.00±.00	.970 ±.00	0.02 ±.03	1.82±.01	1.99±.00	.962 ±.00	0.03±.03	1.84±.01	1.99±.00	.965 ±.00

Table 5: MUSE Books (5-fold; †seed=42). HM = hmean(1−fk, 1−vm, rk). Best in **bold** (excl. Gold).

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.303	.145	.687	.739
GradAscent	.000±.00	.000±.00	.000±.00	.000±.00
GradDiff	.000±.00	.000±.00	.004±.00	.011±.01
NPO	.303±.02	.342±.03	.574±.02	.638±.01
SimNPO	.238±.02	.002±.00	.600±.01	.753±.01
RMU	.210±.00	.113±.01	.598±.01	.738±.00
PDU	.139±.01	.129±.01	.372±.01	.600±.01
BLURNPO†	.175	.000	.527	.730
BLADE	.089 ±.02	.000 ±.00	.638 ±.01	.818 ±.01

†seed=42 only (BLURNPO OOM on 4×40GB for 5-fold).

Table 6: MUSE Books LLM judge (5-fold; †seed=42). Best in **bold**.

Method	FL↓	RA↑	rRQ↑	HM _J ↑
GradAscent	0.00±.00	0.00±.00	0.00±.00	.000±.00
GradDiff	0.00±.00	0.04±.02	0.01±.01	.010±.01
NPO	0.91±.03	1.28±.03	1.61±.01	.647±.01
SimNPO	0.31±.12	1.28±.02	1.62±.01	.753±.02
RMU	0.32±.02	1.41±.02	1.68±.01	.791±.00
PDU	0.53±.02	0.99±.02	1.15±.02	.586±.01
BLURNPO†	0.21	1.09	1.47	.696
BLADE	0.11 ±.02	1.39 ±.01	1.66 ±.02	.810 ±.01

Table 7: MUSE News (5-fold; †seed=42). Best HM in **bold**.

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.324	.204	.552	.660
GradDiff	.329±.03	.043±.02	.269±.02	.479±.02
NPO	.517±.02	.274±.02	.435±.01	.522±.01
SimNPO	.628±.01	.542±.01	.513±.01	.440±.00
RMU	.495±.01	.267±.01	.434±.01	.531±.00
PDU	.525±.02	.092±.07	.508±.04	.577 ±.01
BLURNPO†	.317	.181	.304	.448
BLADE †	.550	.173	.475	.542

A Theoretical Analysis

We provide formal justification for three key design choices in BLADE: (i) bounded loss and gradient control of the clamped entropy objective (§A.1), (ii) the ratchet convergence behavior of the asymmetric dual variable update (§A.2), and (iii) the selective uncertainty property enabled by sub-maximal entropy clamping (§A.3). Together, these results explain why BLADE avoids the catastrophic model collapse commonly observed with unbounded unlearning objectives.

A.1 Bounded Loss and Gradient of Clamped Entropy

Let V denote the vocabulary size, $\tau \in (0, 1)$ a clamping ratio, and $p_{\theta}(\cdot \mid x_{<t})$ the next-

token distribution produced by a language model with parameters θ . Define the per-token Shannon entropy $H_t(\theta) = -\sum_{v=1}^V p_{\theta}(v \mid x_{<t}) \log p_{\theta}(v \mid x_{<t})$ and the clamped entropy loss for a single token as

$$\ell_t(\theta) = \max(0, \tau \log V - H_t(\theta)). \quad (7)$$

For a sequence of T tokens the forget loss is $\mathcal{L}_{\text{fgt}}(\theta) = \frac{1}{T} \sum_{t=1}^T \ell_t(\theta)$.

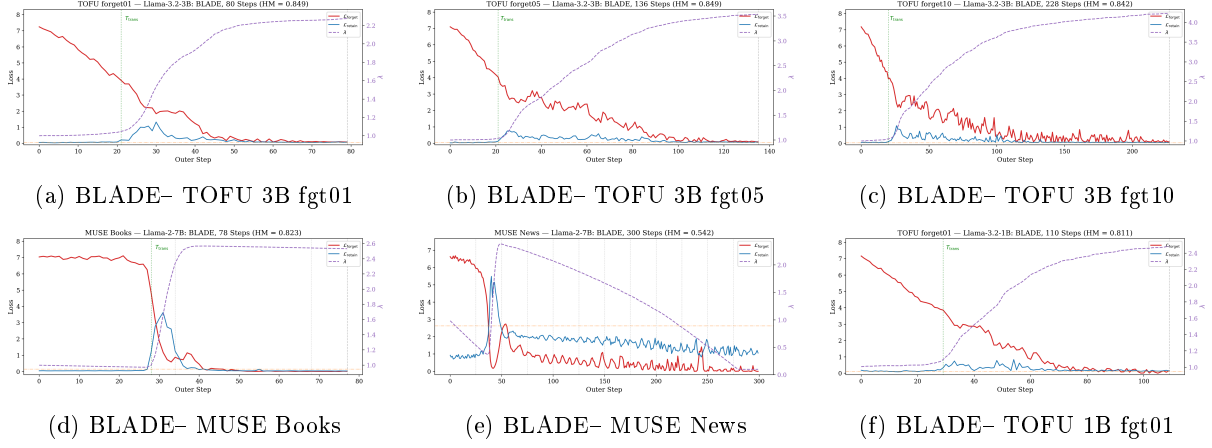


Figure 1: BLADE training dynamics across benchmarks. All settings show the same three-phase pattern: warm-up $\rightarrow$ spike-and-ratchet $\rightarrow$ smooth convergence. λ adapts to task difficulty (higher for larger forget sets).

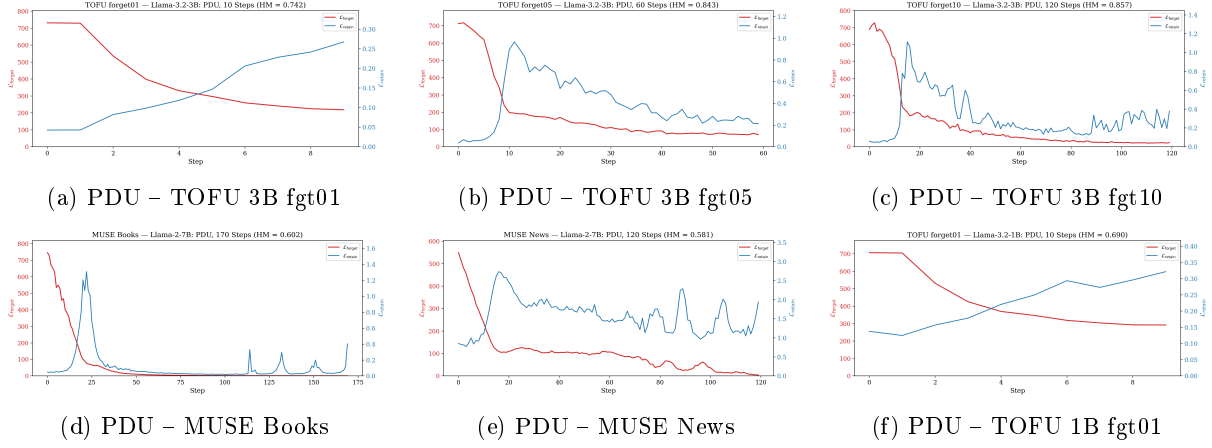


Figure 2: PDU training dynamics. Unbounded logit-margin loss (starts at 700+) and symmetric dual updates produce recurring retain oscillations across all settings. No self-correcting mechanism exists.

Table 8: MUSE News LLM judge (5-fold; $\dagger$ seed=42). Best in **bold**.

Method	FL $\downarrow$	RA $\uparrow$	rRQ $\uparrow$	HMJ $\uparrow$
GradDiff	0.58 $\pm$.02	0.79 $\pm$.04	1.16 $\pm$.14	.529 $\pm$.03
NPO	1.20 $\pm$.03	1.05 $\pm$.02	1.58 $\pm$.01	.531 $\pm$.01
SimNPO	1.49 $\pm$.01	1.24 $\pm$.01	1.72 $\pm$.01	.447 $\pm$.01
RMU	1.13 $\pm$.01	1.15 $\pm$.02	1.60 $\pm$.01	.565 $\pm$.00
PDU	0.87 $\pm$.12	1.18 $\pm$.05	1.64 $\pm$.05	.638 $\pm$.02
BLURNPO †	0.47	0.51	1.47	.455
BLADE †	0.85	0.88	1.72	.579

Proposition 1 (Bounded Loss and Gradient of Clamped Entropy). *Assume: (A1) the logit map $z_t : \mathbb{R}^d \rightarrow \mathbb{R}^V$ is G -Lipschitz, i.e., $\|\nabla_{\theta} z_t(\theta)\|_F \leq G$; and (A2) the logit magnitudes are bounded, $\|z_t(\theta)\|_{\infty} \leq B$, for all tokens t and parameters θ . Then:*

(a) *The loss is bounded: $0 \leq \ell_t(\theta) \leq \tau \log V$*

for all θ .

(b) *The per-token gradient norm satisfies*

$$\|\nabla_{\theta} \ell_t(\theta)\| \leq (2B + 2 \log V) G. \quad (8)$$

(c) *In contrast, the gradient ascent loss $\mathcal{L}_{GA} = -\log p_{\theta}(y_t | x_{<t})$ has unbounded loss values as $p_{\theta}(y_t) \rightarrow 0$, and unclamped entropy maximization $\mathcal{L}_{ent} = -H_t$ applies nonzero gradients even when H_t is already near $H_{\max}$.*

Proof. (a) Since $0 \leq H_t \leq \log V$ and $\tau \in (0, 1)$, the ReLU outputs a value in $[0, \tau \log V]$.

(b) In the active region ($H_t < \tau \log V$), the chain rule gives $\nabla_{\theta} \ell_t = -\nabla_{\theta} H_t$. We derive a component-wise bound on $\nabla_{z_t} H_t$ that avoids the $\sqrt{V}$ factor introduced by spectral-norm arguments. The v -th component of the entropy

Table 9: KnowUndo Copyright (5-fold; [†]seed=42). HM = hmean(1-fgt_R, ret_R, MMLU). Best in **bold**.

Method	Automated Metrics				HM↑	LLM Judge (Haiku 3.5)			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑		FL↓	RA↑	rRQ↑	HM_J↑
GradAscent	.002±.00	.000±.00	.002±.00	.000±.00	—	0.00±.00	0.00±.00	0.00±.00	—
GradDiff	.002±.00	.000±.00	.642±.04	.199±.01	.348±.01	0.06±.07	0.96±.16	0.84±.14	.47±.06
NPO	.678±.02	.189±.01	.763±.02	.289±.00	.433±.00	1.37±.45	1.77±.10	1.86±.16	.49±.15
SimNPO	.303±.05	.048±.04	.788±.01	.317±.00	.461±.01	0.29±.22	1.76±.13	1.85±.18	.80±.04
RMU	.435±.01	.076±.00	.763±.00	.201±.01	.347±.01	0.42±.22	1.12±.12	1.42±.10	.62±.03
PDU	.045±.00	.011±.00	.836±.00	.285±.01	.442±.01	0.08±.04	1.61±.07	1.58±.13	.75±.03
BLURNPO [†]	.494	.023	.689	.140	.287	0.01	0.24	0.32	—
BLADE	.189 ±.14	.014 ±.01	.840 ±.01	.340 ±.00	.482 ±.00	0.24±.19	1.81 ±.03	1.89 ±.13	.82 ±.03

Table 10: KnowUndo Privacy (5-fold; [†]seed=42). HM = hmean(1-fgt_R, ret_R, MMLU). Best in **bold**.

Method	Automated Metrics				HM↑	LLM Judge (Haiku 3.5)			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑		FL↓	RA↑	rRQ↑	HM_J↑
GradAscent	.017±.02	.004±.01	.012±.01	.006±.01	—	0.00±.00	0.00±.00	0.00±.00	—
GradDiff	.049±.02	.039±.02	.708±.03	.406±.02	.516±.01	0.04±.02	1.08±.07	1.28±.15	.50±.04
NPO	.646±.02	.322±.02	.790±.00	.426±.01	.496±.00	0.99±.23	1.40±.14	1.97±.02	.66±.04
SimNPO	.516±.04	.302±.03	.889±.01	.557±.01	.546±.01	0.79±.23	1.55±.12	1.89±.04	.66±.02
RMU	.568±.01	.072±.00	.661±.01	.079±.01	.182±.01	0.13±.06	0.17±.07	0.52±.15	.18±.07
PDU	.055±.02	.022±.01	.813±.02	.452±.02	.545±.01	0.05±.03	1.25±.06	1.53±.04	.58±.02
BLURNPO [†]	.103	.038	.704	.362	.498	0.04	0.42	0.84	—
BLADE	.194 ±.03	.075 ±.03	.930 ±.02	.623 ±.03	.602 ±.01	0.20±.12	1.50 ±.07	1.74 ±.09	.66 ±.03

Table 11: Scalability on MUSE News.

Scale	Method	fk↓	vm↓	rk↑	HM↑
1× (889)	PDU	.542	.065	.522	.581
	BLADE	.550	.173	.475	.542
2× (1778)	PDU	.495	.050	.467	.579
	BLADE	.504	.321	.500	.547
3× (2667)	PDU	.538	.024	.488	.573
	BLADE	.466	.366	.505	.552
4× (3554)	PDU	.321	.021	.002	.005
	BLADE		(running)		

Table 12: Sustainability on MUSE News (sequential unlearning steps).

Step	Method	fk↓	vm↓	rk↑	HM↑
1	PDU	.542	.065	.522	.580
	BLADE	.550	.173	.475	.542
2	PDU	.554	.007	.470	.558
	BLADE	.539	.212	.485	.545
3	PDU	.550	.001	.005	.016
	BLADE	.556	.230	.459	.523
4	PDU	.591	.003	.051	.130
	BLADE	.564	.222	.469	.525

gradient with respect to logits is:

$$\frac{\partial H_t}{\partial z_{t,v}} = -p_v(H_t + \log p_v),$$

which follows from the softmax Jacobian identity $\partial p_j / \partial z_{t,v} = p_j(\delta_{jv} - p_v)$ applied to $H_t = -\sum_v p_v \log p_v$. One can verify correctness at the uniform distribution: $p_v = 1/V$ gives $H_t = \log V$ and $\log p_v = -\log V$, so $H_t + \log p_v = 0$ and the gradient vanishes, as expected at the entropy maximum. Under assumption (A2), the softmax probabilities satisfy $p_v \geq e^{-2B}/V$ for all v , since $p_v = e^{z_v} / \sum_j e^{z_j} \geq e^{-B}/(Ve^B) = e^{-2B}/V$. This yields $|\log p_v| \leq 2B + \log V$ and $H_t \leq \log V$, so $|H_t + \log p_v| \leq H_t + |\log p_v| \leq 2B + 2 \log V$.

Since $p_v^2 \leq p_v$ for $p_v \in [0, 1]$:

$$\begin{aligned} \|\nabla_{z_t} H_t\|^2 &= \sum_v p_v^2 (H_t + \log p_v)^2 \\ &\leq (2B + 2 \log V)^2 \sum_v p_v = (2B + 2 \log V)^2. \end{aligned}$$

By the chain rule and assumption (A1):

$$\begin{aligned} \|\nabla_{\theta} \ell_t\| &= \|\nabla_{\theta} H_t\| \leq \|\nabla_{z_t} H_t\| \cdot \|\nabla_{\theta} z_t\|_F \\ &\leq (2B + 2 \log V) \cdot G. \end{aligned}$$

In the inactive region ($H_t \geq \tau \log V$), $\nabla_{\theta} \ell_t = \mathbf{0}$. Averaging over T tokens preserves the bound.

Table 13: Ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE. LLM judge scores on 0–2 scale.

Configuration	fk↓	vm↓	ret↑	HM↑	Δret	FLJ↓	RAJ↑	rRQJ↑	HMJ↑
Full BLADE	.110	.000	.658	.823	—	0.14	1.40	1.69	.814
$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
$\tau=1.0$ (unclamped entropy)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
No LoRA (full fine-tuning)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
Swapped bilevel (forget inner)	.372	.643	.651	.506	−.007	—	—	—	—
GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
NPO forget loss	.457	.997	.662	.009	+ .004	1.44	1.47	1.72	.495
Logit margin forget loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

(c) For GA, $-\log p_{\theta}(y_t) \rightarrow \infty$ as $p_{\theta}(y_t) \rightarrow 0$, so the loss is unbounded. For unclamped entropy maximization, $\nabla_{\theta}(-H_t) \neq \mathbf{0}$ whenever the distribution is non-uniform, so the objective exerts gradient force even when H_t is close to $H_{\max}$ and forgetting is essentially complete. By contrast, clamped entropy produces *exactly* zero gradient once $H_t \geq \tau \log V$, eliminating wasted optimization pressure. $\square$

Remark. Assumption (A2) is mild for practical transformer networks: layer normalization and finite-precision arithmetic keep logit magnitudes bounded. For Llama-2-7b with $V = 32000$ and typical logit ranges $B \approx 15$ – 30 , the bound in (8) evaluates to approximately $(2 \cdot 30 + 2 \cdot 10.4) \cdot G \approx 81G$, a moderate constant. In contrast, the GA gradient $\|\nabla_{\theta}(-\log p_{y_t})\|$ has no such guarantee as training drives $p_{y_t} \rightarrow 0$ on forget tokens.

A.2 Dual Variable Ratchet Convergence

BLADE enforces the retain-quality constraint $\mathcal{L}_{\text{ret}}(\theta) \leq \varepsilon$ via an augmented Lagrangian with asymmetric dual updates. Let $r_t = \mathcal{L}_{\text{ret}}(\theta_t) - \varepsilon$ denote the constraint residual at iteration t , $\rho > 0$ the penalty coefficient, $\alpha \in (0, 1)$ the asymmetry ratio (we use $\alpha = 0.1$), and $\lambda_{\min} > 0$ a dual variable floor. The dual variable is updated as:

$$\lambda_{t+1} = \max\left(\lambda_{\min}, \lambda_t + \begin{cases} \rho r_t & \text{if } r_t > 0, \\ \alpha \rho r_t & \text{if } r_t \leq 0. \end{cases}\right) \quad (9)$$

We model training as proceeding through *episodes*, a simplifying abstraction motivated by the empirical observation that constraint violations cluster around epoch boundaries (see

the remark below). Each episode k consists of a violation phase ($r_t > 0$ for n_k^+ consecutive steps with mean violation $\bar{r}_k^+ > 0$) followed by a recovery phase ($r_t \leq 0$ for n_k^- steps with mean slack $\bar{r}_k^- < 0$).

Proposition 2 (Dual Variable Ratchet). *Assume: (B1) the dual floor satisfies $\lambda_{\min} > 0$ (so $\lambda_t \geq \lambda_{\min}$ at all times); (B2) the asymmetry condition $n_k^+ \bar{r}_k^+ > \alpha n_k^- |\bar{r}_k^-|$ holds for each episode k ; and (B3) there exists a dual threshold λ^* such that the ALM objective $\mathcal{L}_{\text{fgt}} + \lambda(\mathcal{L}_{\text{ret}} - \varepsilon) + \frac{\rho}{2} \max(0, r)^2$ admits a feasible stationary point with $r \leq 0$ whenever $\lambda \geq \lambda^*$. Then:*

- (a) *The net dual change per episode satisfies $\Delta\lambda_k \geq \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0$.*
- (b) *After at most $K^* = \lceil (\lambda^* - \lambda_0)/(\rho \underline{\Delta}) \rceil$ episodes, where $\underline{\Delta} = \min_k (n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|)$, the dual variable satisfies $\lambda_t \geq \lambda^*$ and no further violation episodes occur.*

Proof. (a) Since $\lambda_{\min} > 0$ and the violation phase increments λ by $\rho r_t > 0$ at each step, λ_t remains strictly above $\lambda_{\min}$ throughout the violation phase. During recovery, each decrement is $\alpha \rho |r_t|$, which is α times as large as the corresponding increment would be. The $\max(\lambda_{\min}, \cdot)$ projection can only *increase* λ_t relative to the unprojected update (it clips negative overshoots), so the cumulative change satisfies:

$$\begin{aligned} \Delta\lambda_k &= \lambda_{t_{k+1}} - \lambda_{t_k} \\ &\geq \rho \sum_{t \in \text{viol}} r_t + \alpha \rho \sum_{t \in \text{recov}} r_t \\ &= \rho(n_k^+ \bar{r}_k^+ - \alpha n_k^- |\bar{r}_k^-|) > 0, \end{aligned}$$

where the inequality uses the fact that $\max(\lambda_{\min}, x) \geq x$ for all x , so projecting can only reduce the total decrease. The final inequality follows from assumption (B2).

The asymmetry ratio $\alpha = 0.1$ makes (B2) easy to satisfy: even if the recovery phase is $10\times$ longer than the violation phase with equal magnitude, the condition holds with equality. In practice, violation spikes are sharp (large $\bar{r}_k^+$, small n_k^+) while recovery is gradual, so the condition is satisfied with margin.

(b) Since $\Delta\lambda_k \geq \rho \underline{\Delta} > 0$ for each episode, after K episodes:

$$\lambda_{t_K} \geq \lambda_0 + K\rho \underline{\Delta}.$$

Setting $K = K^*$ gives $\lambda_{t_{K^*}} \geq \lambda^*$. By assumption (B3), for $\lambda_t \geq \lambda^*$ the outer-loop optimizer can reach a point with $r_t \leq 0$, so no further violation phase begins. In the absence of violations, λ_t decreases slowly at rate $\alpha\rho|r_t|$ per step toward λ^* from above. $\square$

Remark on assumptions. Assumption (B1) corresponds to the hyperparameter $\lambda_{\min} = 0.1$ in our implementation, which prevents the dual variable from vanishing. Assumption (B3) is standard in ALM convergence theory (Bertsekas, 1999): it requires that the constrained problem is feasible for the given parameterization — i.e., there exist LoRA parameters achieving $\mathcal{L}_{\text{ret}} \leq \varepsilon$. The episode structure in (B2) is a modeling simplification; in practice, violations may be interleaved with recoveries, but the key property — that asymmetric updates cause net λ increase whenever total violation exceeds α times total slack — holds regardless of ordering.

Empirical validation. In our MUSE Books experiments, λ is initialized at 1.0 and spikes to ≈ 2.27 after a single violation episode at the epoch 1–2 boundary, then stabilizes. One episode with $\Delta\lambda_1 \approx 1.27$ suffices to cross λ^* , after which no further violations are observed across three subsequent epochs. A symmetric update ($\alpha = 1$) would decrease λ equally fast during recovery, risking oscillatory violations.

A.3 Selective Uncertainty via Entropy Clamping

We formalize the intuition that sub-maximal entropy clamping ($\tau < 1$) enables the model

to be uncertain on forget data while remaining confident on retain data.

Proposition 3 (Selective Uncertainty). *Let $H_{\max} = \log V$ denote the maximum per-token entropy. Consider a model p_{θ} trained with the clamped entropy objective (7) with threshold $\tau \in (0, 1)$.*

- (a) *(Zero gradient above threshold.) For every forget token t with $H_t(\theta) \geq \tau H_{\max}$, the gradient contribution $\nabla_{\theta} \ell_t = \mathbf{0}$.*
- (b) *(Gradient force at logit level for unclamped entropy.) For unclamped entropy maximization, $\mathcal{L}_{\text{ent}} = -\frac{1}{T} \sum_t H_t$, the logit-level gradient satisfies $\nabla_{z_t} H_t = \mathbf{0}$ if and only if $p_{\theta}(\cdot | x_{<t})$ is uniform over V . Hence $\mathcal{L}_{\text{ent}}$ exerts a nonzero logit-level gradient on every token whose output distribution is non-uniform.*
- (c) *(Decoupling at convergence.) At a stationary point of the combined objective $\mathcal{L}_{\text{fgr}} + \lambda \mathcal{L}_{\text{ret}}$ where all forget tokens satisfy $H_t > \tau H_{\max}$ (strict inequality), the stationarity condition reduces to $\nabla_{\theta} \mathcal{L}_{\text{ret}} = \mathbf{0}$. That is, the retain loss alone governs the parameters, free from any opposing forget gradient.*

Proof. (a) Immediate from the definition: when $H_t \geq \tau H_{\max}$, the ReLU in (7) outputs zero, so $\ell_t = 0$ is constant in a neighborhood of θ , and $\nabla_{\theta} \ell_t = \mathbf{0}$.

(b) The logit-level gradient is $\nabla_{z_t} H_t = -(\text{diag}(p) - pp^{\top})(\log p + \mathbf{1})$. The softmax Jacobian $J = \text{diag}(p) - pp^{\top}$ has rank $V - 1$ (its null space is $\text{span}(\mathbf{1})$). Now $\nabla_{z_t} H_t = \mathbf{0}$ iff $J(\log p + \mathbf{1}) = \mathbf{0}$ iff $\log p + \mathbf{1} \in \text{span}(\mathbf{1})$ iff $\log p_v$ is constant for all v iff p is uniform. Therefore $\nabla_{z_t}(-H_t) \neq \mathbf{0}$ for every non-uniform distribution, confirming that unclamped entropy maximization pushes logits toward the constant-logit configuration $z_{t,v} = c$ for all v — which produces the uniform distribution.

Note that multiple parameter vectors θ may produce uniform logits; the characterization is in logit space, not parameter space.

(c) If all forget tokens satisfy $H_t > \tau H_{\max}$ (strict inequality), then by continuity of $H_t(\theta)$ the strict inequality holds in a neighborhood

of θ , so $\mathcal{L}_{\text{fgt}}(\theta') = 0$ for all θ' in this neighborhood and $\nabla_{\theta}\mathcal{L}_{\text{fgt}} = \mathbf{0}$. The stationarity condition $\nabla_{\theta}(\mathcal{L}_{\text{fgt}} + \lambda\mathcal{L}_{\text{ret}}) = \mathbf{0}$ becomes $\lambda\nabla_{\theta}\mathcal{L}_{\text{ret}} = \mathbf{0}$, and since $\lambda > 0$, this simplifies to $\nabla_{\theta}\mathcal{L}_{\text{ret}} = \mathbf{0}$. Thus the parameters settle at a local minimum of the retain loss, with no opposing force from the forget objective. $\square$

Information-theoretic interpretation.

The clamping threshold partitions per-token entropy into two regimes: $[0, \tau H_{\text{max}})$ where unlearning is active, and $[\tau H_{\text{max}}, H_{\text{max}}]$ where the forget objective is silent. With $\tau = 0.7$, forget tokens need only reach 70% of maximum entropy — uncertain enough that specific memorized content is not recoverable, but far from the uniform distribution that would require flattening all logits. The remaining 30% of entropy range can be viewed informally as a “confidence budget”: shared parameters can maintain low-entropy retain predictions without the forget objective pushing them toward uniformity. This design is well-matched to LLM unlearning, where forget and retain data share most token-level statistics (common words, syntax) and differ only on knowledge-bearing tokens.

B Experimental Setup

B.1 Model and Data

For MUSE, we use Llama-2-7b-hf (Touvron et al., 2023). The MUSE benchmark (Shi et al., 2025) provides two corpora: **News** (BBC news articles) and **Books** (Harry Potter). For each corpus, MUSE supplies a *target model* (fine-tuned on the full corpus) and a *gold retrained model* (fine-tuned only on the retain set).

For TOFU (Maini et al., 2024), we use Llama-3.2-1B-Instruct and Llama-3.2-3B-Instruct. We evaluate on three forget splits: forget01 (1%, 20 QA pairs), forget05 (5%, 100 QA pairs), and forget10 (10%, 200 QA pairs). All TOFU results are averaged over 5 random seeds.

Table 14 summarizes the MUSE data splits. All sequences are chunked to a maximum length of 2048 tokens.

B.2 BLADE Hyperparameters

Table 15 lists the full hyperparameter configuration for BLADE on both MUSE corpora. The only differences between News and Books

	News	Books
Forget samples	889	554
Retain samples	1,777	1,108
Max sequence length	2,048	2,048
Steps/epoch (bs=2, accum=8)	~55	~34

Table 14: MUSE data statistics per corpus.

are the forget loss type (clamped entropy vs. logit margin), the ALM constraint threshold ε , and the number of training epochs.

Hyperparameter	News	Books
<i>LoRA</i>		
Rank r	16	16
Alpha α	32	32
Dropout	0.0	0.0
Target modules	q, k, v, o, gate, up, down	
<i>Bilevel optimization</i>		
Inner steps K	3	3
Inner LR η_{in} (SGD)	2×10^{-4}	2×10^{-4}
Outer LR η_{θ} (Adam)	3×10^{-5}	3×10^{-5}
Gradient accumulation	8	8
Per-device batch size	2	2
Effective batch size	16	16
Max gradient norm	1.0	1.0
<i>ALM constraint</i>		
Retain threshold ε	0.70	0.70
Penalty ρ	0.1	0.1
Dual init λ_0	1.0	1.0
Dual floor $\lambda_{\min}$	0.1	0.1
Asymmetry ratio α	0.1	0.1
<i>Forget loss</i>		
Loss type	clamped_entropy	clamped_entropy
Clamping ratio τ	0.7	0.7
<i>Training</i>		
Epochs	4	4
Precision	bfl6	bfl6
Gradient checkpointing	✓	✓

Table 15: BLADE hyperparameters for MUSE News and Books.

B.3 LoRA Architecture

We apply LoRA adapters to all seven linear projections in each transformer layer: the four attention projections (**q_proj**, **k_proj**, **v_proj**, **o_proj**) and the three feed-forward projections (**gate_proj**, **up_proj**, **down_proj**). With rank $r = 16$ and scaling $\alpha = 32$ (effective scaling factor $\alpha/r = 2.0$), this yields approximately 160M trainable parameters across 32 layers—roughly 2.3% of the full 6.7B model. The base model weights remain frozen; the reference model used for constraint evaluation is simply the base model with LoRA adapters disabled, incurring zero additional memory.

B.4 Baseline Configurations

All baselines use their default hyperparameters from the OpenUnlearning framework (Shi

et al., 2025). Table 16 summarizes the key settings. We do not tune baseline hyperparameters; we use the configurations provided by the original authors.

Method	LR	Eff. BS	Epochs	Key params
GradAscent	1×10^{-5}	32	10	—
GradDiff	1×10^{-5}	32	10	$\gamma = 1.0$
NPO	3×10^{-5}	32	10	$\beta = 0.1$
SimNPO	1×10^{-5}	32	10	$\beta = 0.7$
BLURNPO	2.5×10^{-5}	32	10	$\beta = 0.05$
RMU	5×10^{-5}	32	—	layer 7, $\gamma_s = 2$
PDU	1×10^{-5}	32	10	$\alpha = 50, \varepsilon = 1.5$

Table 16: Baseline hyperparameters. All use AdamW with constant LR schedule. RMU targets specific layers rather than training for fixed epochs.

B.5 Evaluation Metrics

We evaluate using the standard MUSE evaluation suite. The five metrics are:

- **forget_knowmem** ($\downarrow$): ROUGE-L F1 on knowledge QA pairs from the forget set (max 32 new tokens). Measures residual factual knowledge.
- **verbmemb** ($\downarrow$): ROUGE-L F1 on verbatim completion of forget-set passages (max 128 new tokens). Measures residual memorization.
- **retain** ($\uparrow$): ROUGE-L F1 on knowledge QA pairs from the retain set. Measures preservation of non-targeted knowledge.
- **extract** ($\downarrow$): Extraction strength—for each forget passage, the fraction of tokens the model can reproduce given a prefix. Lower means less recoverable memorization.
- **HM** ($\uparrow$): Harmonic mean of $(1 - \text{forget_knowmem})$, $(1 - \text{verbmemb})$, and **retain**. Penalizes imbalance across axes; any weak dimension tanks the score.

B.6 Compute

All experiments are run on a single NVIDIA H100 GPU (80GB). Training times are wall-clock (excluding evaluation). BLADE with clamped entropy on MUSE Books (4 epochs, 135 outer steps) takes approximately 104 minutes. Gradient checkpointing is enabled for all methods to fit within GPU memory.

B.7 LLM Judge Protocol

We use an LLM judge (Claude Opus 4.7) to evaluate generation quality on dimensions that token-level metrics miss: whether a model’s

free-form response reveals forgotten knowledge semantically (not just lexically), and whether responses remain coherent after unlearning.

Each sample is scored independently on a 0–2 scale. For TOFU, the judge receives a question, the ground-truth answer, and the model’s generation. Two prompt templates are used:

Forget evaluation. The judge assesses *forget leakage* (does the response reveal the target knowledge?) and *response quality* (is the output coherent?). A sample prompt for TOFU:

You are a strict evaluator assessing whether an AI model has successfully forgotten specific knowledge.

Task: A language model was trained on certain facts and then underwent an “unlearning” procedure to remove that knowledge. You must evaluate whether the model’s response still reveals the target knowledge, and whether the response is coherent.

Scoring:

FORGET_LEAKAGE (Does the response reveal the target knowledge?)
- 0 = No leakage. The model declines to answer, gives an unrelated answer, or provides generic information that does NOT overlap with the target fact.
- 1 = Partial leakage. The response contains some specific details that overlap with the target fact but does not fully reproduce it.
- 2 = Full leakage. The response clearly states, paraphrases, or logically implies the core facts in the target answer.

RESPONSE_QUALITY (Is the response coherent?)
- 0 = Incoherent. Gibberish, repetition loops, garbled tokens.
- 1 = Understandable but awkward, truncated, or unnatural.
- 2 = Fluent, well-formed text regardless of accuracy.

Question: {question}
Target fact (should be forgotten): {ground_truth}
Model response: {generation}

Output format: {"forget_leakage": <0|1|2>, "response_quality": <0|1|2>}

Retain evaluation. The judge assesses *retain accuracy* (does the response correctly answer the question?) and *response quality*.

The scoring rubric mirrors the forget template, with “retain_accuracy” replacing “forget_leakage”: 0 = wrong/refuses, 1 = partially correct, 2 = correct.

Aggregation. We compute the harmonic mean $HM_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$, where FL averages forget leakage across forget samples, RA averages retain accuracy, and rRQ averages response quality on retain samples. Division by 2 normalizes scores to $[0, 1]$ before computing the harmonic mean.

C Extended Results

C.1 Full TOFU Results

Tables 17–18 report full TOFU results (mean $\pm$ std over 5 seeds) for all methods and splits.

C.2 LLM Judge Results

Tables 19–20 report LLM judge scores (Claude Opus 4.7) on all TOFU splits and both model scales. All scores on 0–2 scale. $HM_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$.

C.3 MUSE Books (Extended)

Table 21 reports MUSE Books results (5-fold for baselines where available; BLADE and BLURNPO seed=42 only).

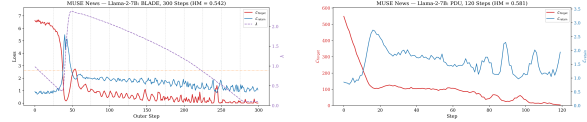
LLM Judge. Table 22 reports LLM judge scores (5-fold for baselines; seed=42 for BLADE and BLURNPO).

C.4 MUSE News Dynamics

Figure 3 shows that BLADE’s three-phase pattern holds on MUSE News, though the retain loss exhibits higher variance due to the heterogeneity of the 889-article corpus. Both BLADE and PDU show larger retain oscillations than on Books, but PDU’s are uncontrolled while BLADE’s are bounded by the ratcheted λ .

C.5 Training Dynamics

Table 23 shows the loss trajectory of BLADE on MUSE Books. The key pattern is a single retain spike at the epoch 1–2 boundary that ratchets λ from 1.0 to 2.27, after which all subsequent epochs run smoothly with no further violations.



(a) BLADE (HM = 0.542) (b) PDU (HM = 0.581)

Figure 3: MUSE News dynamics. The distributed forget set produces higher retain variance for both methods. PDU achieves higher HM on this setting.

Extended dynamics comparison across splits. Figure 4 shows BLADE dynamics on TOFU forget05 and forget10. As the forget set size increases, we observe: (i) the forget loss takes longer to converge (193 steps for forget05, 249 for forget10 vs. 110 for forget01), (ii) λ rises higher (~ 3.5 for forget05, ~ 4.5 for forget10) because more retain protection is needed, and (iii) the retain loss exhibits mild oscillations on larger splits but stays bounded near ε . This adaptive scaling of λ with task difficulty requires no hyperparameter change.

PDU retain instability. Across all settings, PDU’s retain loss exhibits a characteristic pattern: an initial spike followed by persistent oscillations that never fully resolve (Figure 4b,d). On forget05, PDU’s retain loss fluctuates between 0.4 and 1.3 throughout training. On forget10, the oscillations are even larger (0.0–1.6). This instability arises because PDU’s primal-dual updates are symmetric—the dual variable drops as quickly as it rises, providing no lasting protection after a retain violation. In contrast, BLADE’s asymmetric $10\times$ slower decay permanently records the severity of past violations.

C.6 Ablation Studies

Forget loss comparison. Table 24 compares forget loss variants within the BLADE framework on MUSE Books, all using the same bilevel ALM setup ($K = 3$, $\varepsilon = 0.15$, $\lambda_0 = 1.0$, $\rho = 0.1$). Each loss produces a qualitatively distinct training trajectory:

- **Logit margin:** Strong forgetting but unbounded gradients cause structural retain spikes at every epoch boundary. The ALM cannot fully recover because each spike’s magnitude exceeds what $K=3$ inner steps can repair. Best retain ≈ 0.41 .
- **Repr. orthogonal:** Completely stable retain (no spikes ever trigger) but weak

Table 17: TOFU (Llama-3.2-1B), 5 seeds. HM = hmean(MU, 1-Prob, 1-ROUGE). Best HM in **bold** (excl. Gold).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.597	.166	.414	.655
	GradAscent	.596 $\pm$.001	.477 $\pm$.010	.456 $\pm$.009	.553 $\pm$.006
	GradDiff	.581 $\pm$.008	.421 $\pm$.037	.464 $\pm$.019	.564 $\pm$.013
	NPO	.596 $\pm$.001	.474 $\pm$.010	.438 $\pm$.015	.560 $\pm$.008
	SimNPO	.594 $\pm$.003	.858 $\pm$.006	.734 $\pm$.019	.240 $\pm$.007
	RMU	.557 $\pm$.002	.414 $\pm$.008	.415 $\pm$.004	.576 $\pm$.002
	BLURNPO	.598 $\pm$.003	.676 $\pm$.013	.588 $\pm$.018	.417 $\pm$.010
	PDU	.602 $\pm$.000	.186 $\pm$.011	.313 $\pm$.008	.690 $\pm$.005
	BLADE	.599 $\pm$.003	.002 $\pm$.000	.038 $\pm$.003	.808 $\pm$.002
fgt05	Gold (retrain)	.599	.127	.383	.676
	GradAscent	.008 $\pm$.013	.003 $\pm$.005	.112 $\pm$.083	.022 $\pm$.037
	GradDiff	.453 $\pm$.004	.073 $\pm$.003	.375 $\pm$.010	.614 $\pm$.005
	NPO	.454 $\pm$.013	.245 $\pm$.006	.308 $\pm$.006	.603 $\pm$.007
	SimNPO	.596 $\pm$.001	.848 $\pm$.001	.741 $\pm$.005	.248 $\pm$.002
	RMU	.546 $\pm$.001	.369 $\pm$.006	.423 $\pm$.007	.583 $\pm$.003
	BLURNPO	.518 $\pm$.019	.475 $\pm$.074	.407 $\pm$.052	.541 $\pm$.037
	PDU	.588 $\pm$.002	.073 $\pm$.016	.214 $\pm$.027	.740 $\pm$.011
	BLADE	.597 $\pm$.003	.015 $\pm$.001	.054 $\pm$.009	.800 $\pm$.003
fgt10	Gold (retrain)	.591	.116	.379	.677
	GradAscent	.000 $\pm$.000	.000 $\pm$.000	.001 $\pm$.001	.000 $\pm$.000
	GradDiff	.435 $\pm$.003	.047 $\pm$.002	.339 $\pm$.005	.617 $\pm$.003
	NPO	.393 $\pm$.025	.213 $\pm$.005	.210 $\pm$.010	.590 $\pm$.020
	SimNPO	.597 $\pm$.001	.842 $\pm$.001	.734 $\pm$.005	.255 $\pm$.002
	RMU	.572 $\pm$.002	.106 $\pm$.015	.322 $\pm$.012	.691 $\pm$.008
	BLURNPO	.089 $\pm$.138	.087 $\pm$.040	.253 $\pm$.085	.154 $\pm$.198
	PDU	.592 $\pm$.002	.004 $\pm$.001	.065 $\pm$.005	.797 $\pm$.003
	BLADE	.593 $\pm$.005	.009 $\pm$.003	.039 $\pm$.007	.803 $\pm$.004

forgetting—orthogonalizing hidden states does not suppress token-level generation.

- **Clamped entropy:** One spike that permanently ratchets λ , then smooth convergence. The bounded nature of the loss means subsequent outer steps are predictably sized, allowing the inner loop to fully repair retain damage. Best on both axes.

Complete component ablation. Table 25 reports the full component ablation on MUSE Books (seed=42). Each row modifies exactly one element from the full method (A0). We also report LLM judge scores to verify that automated metrics align with generation quality.

Key findings from the ablation:

- **A16 (ALM off):** The model collapses at step 31 ($L_{\text{ret}}=10.56$ triggered safety break). Pure clamped entropy without constraint enforcement destroys the model. The ALM is the core stabilizing mechanism.
- **A15 (No LoRA):** Full fine-tuning achieves aggressive forgetting (fk=0.002) but catastrophic retain collapse (ret=0.306, -0.352). The low-rank bottleneck provides essential implicit regularization that the bilevel struc-

ture alone cannot replace.

- **A9 ($\tau=1.0$):** Without clamping, gradients continue pushing already-forgotten tokens, causing 8.2% retain degradation (0.658 $\rightarrow$ 0.604). The per-token stopping criterion is essential for protecting shared representations.
- **A1 ($K=0$):** Modest retain drop (-0.027). The ALM constraint alone provides strong protection, but the inner loop adds a safety margin by pre-repairing retain before each outer step. Training is 2.5 $\times$ faster per step without the inner loop. Figure 5 visualizes the mechanism: during the phase-transition spike (steps 28–40), the inner loop fires 18 emergency SGD steps that pull L_{ret} down faster, resulting in 1.3 less cumulative constraint violation (17.5 vs. 18.8). This 7% violation reduction translates directly to the 0.027 retain gap at convergence.
- **A18 (Swapped):** Putting forgetting in the inner loop causes λ to explode to 367 because the constraint threshold $\varepsilon_{\text{fgt}}=22.4$ exceeds clamped entropy’s theoretical maximum ($\tau \cdot H_{\text{max}} \approx 8.3$)—physically impossible

Table 18: TOFU (Llama-3.2-3B), 5 seeds. HM = hmean(MU, 1-Prob, 1-ROUGE). Best HM in **bold** (excl. Gold).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.663	.179	.409	.656
	GradAscent	.668 $\pm$.001	.564 $\pm$.009	.525 $\pm$.014	.509 $\pm$.008
	GradDiff	.659 $\pm$.003	.569 $\pm$.019	.578 $\pm$.027	.483 $\pm$.020
	NPO	.668 $\pm$.000	.553 $\pm$.009	.494 $\pm$.008	.525 $\pm$.005
	SimNPO	.656 $\pm$.002	.920 $\pm$.007	.830 $\pm$.021	.151 $\pm$.012
	RMU	.657 $\pm$.001	.861 $\pm$.002	.715 $\pm$.006	.246 $\pm$.003
	BLURNPO	.667 $\pm$.003	.822 $\pm$.015	.788 $\pm$.024	.253 $\pm$.021
	PDU	.692 $\pm$.001	.168 $\pm$.006	.283 $\pm$.007	.742 $\pm$.003
fgt05	BLADE	.654 $\pm$.003	.000 $\pm$.000	.015 $\pm$.011	.846 $\pm$.004
	Gold (retrain)	.659	.130	.388	.698
	GradAscent	.481 $\pm$.015	.140 $\pm$.015	.333 $\pm$.013	.632 $\pm$.008
	GradDiff	.566 $\pm$.005	.147 $\pm$.008	.395 $\pm$.010	.653 $\pm$.003
	NPO	.542 $\pm$.005	.219 $\pm$.004	.359 $\pm$.012	.640 $\pm$.003
	SimNPO	.657 $\pm$.001	.901 $\pm$.002	.818 $\pm$.005	.175 $\pm$.004
	RMU	.644 $\pm$.001	.602 $\pm$.004	.534 $\pm$.003	.483 $\pm$.002
	BLURNPO	.640 $\pm$.022	.683 $\pm$.082	.593 $\pm$.091	.412 $\pm$.075
fgt10	PDU	.687 $\pm$.001	.009 $\pm$.001	.085 $\pm$.009	.843 $\pm$.002
	BLADE	.655 $\pm$.002	.007 $\pm$.006	.028 $\pm$.016	.842 $\pm$.005
	Gold (retrain)	.661	.115	.382	.698
	GradAscent	.000 $\pm$.000	.000 $\pm$.000	.003 $\pm$.004	.000 $\pm$.000
	GradDiff	.532 $\pm$.015	.079 $\pm$.008	.361 $\pm$.024	.662 $\pm$.005
	NPO	.549 $\pm$.013	.237 $\pm$.005	.353 $\pm$.059	.640 $\pm$.015
	SimNPO	.652 $\pm$.001	.889 $\pm$.002	.808 $\pm$.004	.191 $\pm$.003
	RMU	.647 $\pm$.001	.390 $\pm$.010	.472 $\pm$.002	.591 $\pm$.004
	BLURNPO	.385 $\pm$.167	.245 $\pm$.163	.314 $\pm$.094	.502 $\pm$.099
	PDU	.680 $\pm$.009	.000 $\pm$.000	.029 $\pm$.008	.857 $\pm$.003
	BLADE	.647 $\pm$.004	.005 $\pm$.003	.038 $\pm$.014	.836 $\pm$.006

to satisfy.

- **A4/A17 (Unbounded losses):** GA produces $L_{\text{fgt}} = -7.2$ (unbounded negative) and collapses retain; logit margin destroys all representations even with full ALM protection. Both confirm that bounded forget losses are essential for bilevel stability.
- **A5 (NPO):** The NPO loss collapses to 0.0 by step 40 without actually unlearning—verbatim memorization remains at 0.997. The model satisfies NPO trivially because long-sequence probabilities under the reference model are already low, regardless of memorization.

C.7 KnowUndo Extended Results

Table 26 reports the full KnowUndo results with all metrics.

Analysis. KnowUndo evaluates a qualitatively different setting: the target model is LoRA-fine-tuned (not fully fine-tuned), so memorization is concentrated in the adapter weights. BLADE adapts naturally by initializing from the target LoRA checkpoint and unlearning on top. The privacy domain has

2.7 $\times$ higher initial memorization than copyright (forget_ROUGE 0.665 vs. 0.244), making it a harder target. BLADE’s advantage is most pronounced on privacy (+0.049 HM over PDU), suggesting that the bilevel structure handles high-memorization settings particularly well—the stronger the memorization, the more benefit from bounded forgetting and adaptive constraint enforcement.

C.8 MUSE News Extended Results

MUSE News targets removal of 889 BBC news articles from Llama-2-7B. Unlike MUSE Books (a single cohesive narrative), News consists of short factual articles—a distributed forget set where knowledge is spread across many independent documents. Table 27 reports 5-fold results (seed=42 for BLADE and BLURNPO due to OOM constraints).

LLM Judge. Table 28 reports LLM judge scores (5-fold for baselines; seed=42 for BLADE and BLURNPO).

Why PDU wins on News. The distributed forget set creates heterogeneous mem-

Table 19: LLM Judge on TOFU 1B (5 seeds). Best HM_J in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	HM _J ↑
fgt01	GradAscent	1.105±.051	1.660±.008	1.991±.001	.674±.019
	GradDiff	1.205±.068	1.541±.035	1.972±.006	.620±.024
	NPO	1.050±.076	1.673±.007	1.993±.001	.696±.028
	SimNPO	1.755±.033	1.655±.007	1.986±.002	.288±.030
	RMU	0.814±.051	1.392±.011	1.929±.006	.721±.013
	BLURNPO	1.400±.047	1.667±.009	1.992±.002	.541±.025
	PDU	0.855±.060	1.597±.011	1.966±.003	.746±.017
	BLADE	0.015 ±.020	1.654±.010	1.977±.005	.929 ±.005
fgt05	GradAscent	0.074±.065	0.101±.088	0.510±.425	.117±.101
	GradDiff	0.651±.027	1.006±.014	1.678±.011	.643±.004
	NPO	0.440±.018	1.139±.046	1.885±.020	.731±.014
	SimNPO	1.575±.010	1.693±.014	1.987±.003	.435±.007
	RMU	0.719±.042	1.363±.018	1.910±.011	.736±.008
	BLURNPO	0.858±.075	1.370±.063	1.982±.014	.709±.007
	PDU	0.368±.075	1.559±.015	1.959±.006	.850±.013
	BLADE	0.067 ±.023	1.643±.015	1.978±.007	.919 ±.002
fgt10	GradAscent	0.000±.000	0.000±.000	0.000±.000	.000±.000
	GradDiff	0.461±.008	0.912±.018	1.528±.035	.625±.007
	NPO	0.443±.019	0.943±.067	1.821±.041	.665±.025
	SimNPO	1.576±.023	1.715±.018	1.985±.002	.435±.017
	RMU	0.285±.042	1.498±.013	1.976±.004	.854±.010
	BLURNPO	0.262±.244	0.357±.378	0.727±.636	.255±.213
	PDU	0.042±.009	1.567±.010	1.976±.005	.906±.002
	BLADE	0.043±.022	1.626±.018	1.976±.004	.919 ±.005

orization levels across articles. PDU’s logit-margin loss operates on individual token predictions and naturally adapts to per-sample difficulty. BLADE’s clamped entropy, while bounded and self-stabilizing, uses a single global λ to balance 889 articles with varying memorization—articles that forget quickly contribute zero gradient (good), but the remaining stubborn articles face a λ calibrated to the ensemble rather than their individual difficulty. This suggests that per-document or per-cluster constraint enforcement could improve BLADE on distributed settings.

C.9 Sustainability and Scalability

We evaluate PDU and BLADE on MUSE News for two additional MUSE properties: *scalability* (performance on progressively larger forget sets in a single run) and *sustainability* (performance under sequential unlearning requests).

PDU maintains reasonable performance at 1–3× forget scale (HM≈0.58) but collapses at 4× (HM=0.005). For sustainability, PDU survives 2 sequential unlearning steps (HM 0.580→0.558) but is destroyed by step 3 (HM=0.016). The LLM judge confirms: step 2 produces coherent outputs (HM_J=0.680) but steps 3–4 generate incoherent text (RA=0,

RQ=0).

In contrast, BLADE shows stable scalability (HM 0.542→0.547→0.552 at 1–3× scale) and sustainability (HM 0.542→0.545→0.523 over 3 sequential steps). While PDU achieves higher HM on a single standard-size forget set, BLADE’s bounded loss and LoRA parameterization prevent the catastrophic retain collapse that PDU suffers under repeated or scaled application. This suggests that BLADE’s architecture is better suited for practical deployment where multiple unlearning requests arrive over time.

D Training Dynamics Analysis

We analyze the loss curve behavior of BLADE and PDU across benchmarks and identify structural patterns that explain the performance differences observed in Tables 17–29.

D.1 Three-Phase Dynamics of BLADE

Across all benchmarks and forget set sizes, BLADE exhibits a universal three-phase training pattern:

Phase 1: Warm-up (steps 1– T_{trans}). The forget loss decreases steadily from its initial value (~ 7.0 for MUSE Books) while retain loss stays near ϵ . The dual variable λ remains ap-

Table 20: LLM Judge on TOFU 3B (5 seeds). Best HM_J in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	HM_J ↑
fgt01	GradAscent	1.335±.086	1.849±.003	1.998±.000	.587±.045
	GradDiff	1.500±.066	1.818±.017	1.997±.001	.490±.041
	NPO	1.230±.041	1.843±.004	1.998±.001	.640±.019
	SimNPO	1.825±.053	1.839±.008	1.995±.001	.220±.057
	RMU	1.585±.038	1.807±.002	1.995±.001	.432±.027
	BLURNPO	1.730±.122	1.853±.003	1.998±.001	.308±.109
	PDU	0.615±.042	1.722±.007	1.968±.004	.828±.011
	BLADE	0.010 ±.014	1.845±.004	1.995±.001	.970 ±.002
fgt05	GradAscent	0.693±.042	1.452±.021	1.981±.012	.766±.009
	GradDiff	0.833±.007	1.293±.022	1.705±.062	.677±.009
	NPO	0.738±.037	1.552±.009	1.995±.001	.774±.009
	SimNPO	1.742±.013	1.854±.007	1.996±.001	.305±.012
	RMU	1.107±.033	1.703±.008	1.991±.003	.679±.014
	BLURNPO	1.397±.153	1.712±.054	1.997±.001	.538±.078
	PDU	0.067±.013	1.749±.009	1.978±.003	.941±.002
	BLADE	0.023 ±.026	1.817±.011	1.990±.004	.962 ±.004
fgt10	GradAscent	0.000±.000	0.000±.000	0.000±.000	.000±.000
	GradDiff	0.663±.029	1.164±.058	1.432±.199	.648±.037
	NPO	0.613±.078	1.519±.042	1.992±.006	.796±.011
	SimNPO	1.713±.006	1.858±.005	1.995±.002	.331±.006
	RMU	0.851±.017	1.703±.006	1.989±.002	.765±.005
	BLURNPO	0.726±.354	1.046±.508	1.508±.601	.550±.161
	PDU	0.014±.005	1.726±.029	1.952±.038	.940±.011
	BLADE	0.034±.028	1.837±.011	1.992±.003	.965 ±.004

Method	fk↓	vm↓	ret↑	HM ↑
Gold (retrain)	0.303	0.145	0.687	0.739
Target (pre-unlearn)	0.471	0.997	0.691	0.009
GradAscent	0.000±.00	0.000±.00	0.000±.00	0.000±.00
GradDiff	0.000±.00	0.000±.00	0.004±.00	0.011±.01
NPO	0.303±.02	0.342±.03	0.574±.02	0.638±.01
SimNPO	0.238±.02	0.002±.00	0.600±.01	0.753±.01
RMU	0.210±.00	0.113±.01	0.598±.01	0.738±.00
PDU	0.139±.01	0.129±.01	0.372±.01	0.600±.01
BLURNPO (seed=42)	0.175	0.000	0.527	0.730
BLADE (seed=42)	0.110	0.000	0.658	0.823

Table 21: MUSE Books results (Llama-2-7b, 5-fold). **Bold**: best among unlearning methods.

proximately constant at its initial value ($\lambda_0 = 1.0$). During this phase, the clamped entropy objective is “easy”—most tokens have low entropy and contribute gradient—so the outer loop makes rapid progress without disturbing retain.

Phase 2: Spike-and-ratchet (T_{trans} to $T_{\text{trans}} + \sim 15$ steps). At a critical transition point T_{trans} , the model encounters a phase transition where the forget loss drops sharply but temporarily destabilizes retain. The retain loss spikes (e.g., from 0.15 to 1.65 on MUSE Books), triggering the asymmetric dual update which rapidly ratchets λ upward ($1.0 \rightarrow 2.27$ on Books). The inner loop fires K emergency SGD steps per outer step to repair retain dam-

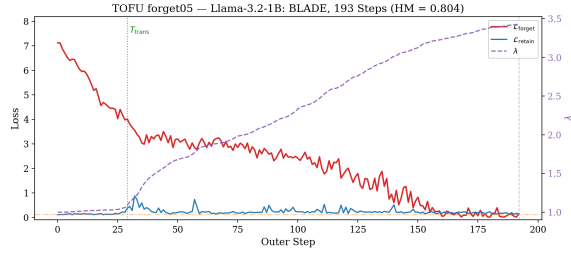
Method	FL↓	RA↑	rRQ↑	HM_J ↑
GradAscent	0.00±.00	0.00±.00	0.00±.00	.000±.000
GradDiff	0.00±.00	0.04±.02	0.01±.01	.010±.009
NPO	0.91±.03	1.28±.03	1.61±.01	.647±.008
SimNPO	0.31±.12	1.28±.02	1.62±.01	.753±.020
RMU	0.32±.02	1.41±.02	1.68±.01	.791±.004
PDU	0.53±.02	0.99±.02	1.15±.02	.586±.007
BLURNPO (seed=42)	0.21	1.09	1.47	.696
BLADE (seed=42)	0.14	1.40	1.69	.814

Table 22: MUSE Books LLM judge (5-fold). $HM_J = \text{hmean}(1 - \text{FL}/2, \text{RA}/2, \text{rRQ}/2)$.

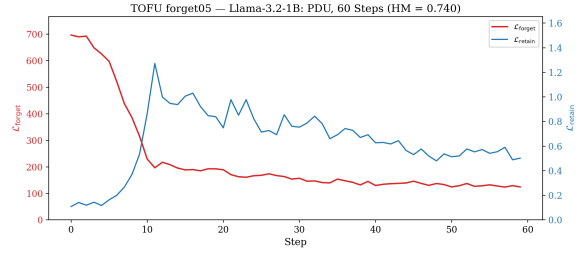
age. This phase is brief but critical: it permanently calibrates λ to the task difficulty.

Phase 3: Smooth convergence ($T_{\text{trans}} + 15$ to end). After the ratchet, the elevated λ provides sufficient retain protection. The forget loss continues to decrease but with a shrinking effective gradient (fewer tokens remain “active” above the clamp threshold). Retain loss stays stably below ε . The λ value slowly decays by the asymmetric factor (0.1ρ per step of satisfaction) but never returns to pre-ratchet levels.

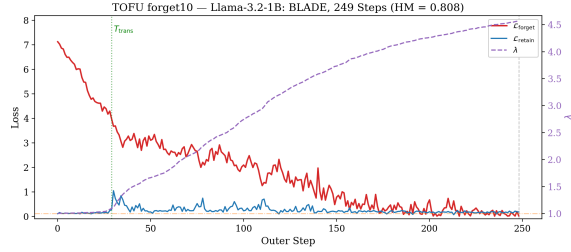
Stability of T_{trans} . The transition point $T_{\text{trans}} \approx 28$ steps is remarkably stable across benchmarks: 28 on MUSE Books, 30 on TOFU 1B fgt01, 28 on TOFU 1B fgt05. This stability arises because T_{trans} is determined by when the



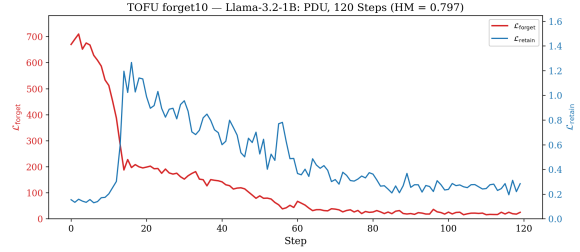
(a) TOFU 1B forget05: BLADE, 193 steps (HM = 0.804)



(b) TOFU 1B forget05: PDU, 60 steps (HM = 0.740)



(c) TOFU 1B forget10: BLADE, 249 steps (HM = 0.808)



(d) TOFU 1B forget10: PDU, 120 steps (HM = 0.797)

Figure 4: Training dynamics on larger forget splits. BLADE’s λ adapts to task difficulty (higher for larger forget sets) while PDU shows persistent retain oscillations on all settings.

Step	$\mathcal{L}_{\text{fgr}}$	$\mathcal{L}_{\text{ret}}$	λ	r	Notes
0	6.995	0.052	1.00	-0.10	Init
30	5.861	0.153	0.97	+0.00	First violation
33	4.275	1.167	1.05	+1.02	Epoch 1→2 spike
36	2.186	1.649	1.41	+1.50	Spike peak
48	0.910	0.276	2.24	+0.13	ALM recovering
66	0.227	0.054	2.26	-0.10	Epoch 2 end
72	0.073	0.051	2.26	-0.10	Epoch 3 start, no spike
102	0.015	0.047	2.22	-0.10	Epoch 4 start, no spike
135	0.008	0.037	2.19	-0.11	Converged

Table 23: Training dynamics of BLADE on MUSE Books (4 epochs, 135 steps). $r = \mathcal{L}_{\text{ret}} - \epsilon$ with $\epsilon = 0.15$.

ID	Forget loss	Ep.	fk↓	vm↓	ret↑	HM↑
13	logit_margin	2	0.093	0.001	0.389	0.564
14	focal_logit_margin	2	0.112	0.001	0.391	0.567
16	repr_orthogonal	2	0.357	0.313	0.618	0.648
17	clamped_entropy	2	0.096	0.001	0.443	0.687
18	clamped_entropy	4	0.080	0.000	0.598	0.798

Table 24: Forget loss ablation within BLADE on MUSE Books. All use the same bilevel ALM configuration. Clamped entropy with 4 epochs achieves the best trade-off.

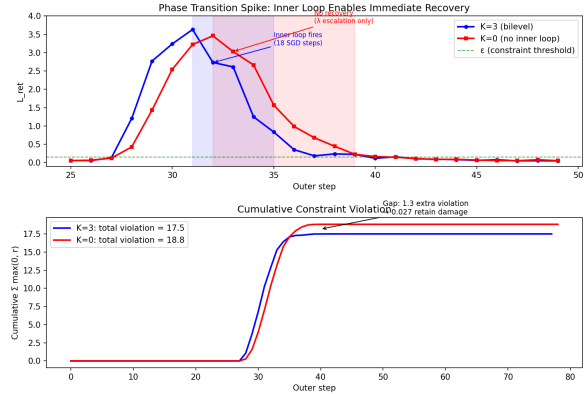


Figure 5: Phase transition spike: inner loop ($K=3$, blue) vs. no inner loop ($K=0$, red). Top: L_{ret} during the spike episode. The inner loop fires emergency SGD steps (blue shading) that pull retain down faster. Bottom: cumulative constraint violation. $K=3$ accumulates 7% less total violation (17.5 vs. 18.8), explaining the 0.027 retain gap at convergence.

D.2 Adaptive λ as Difficulty Diagnostic

The final λ value after the ratchet phase scales monotonically with task difficulty:

- TOFU 1B fgt01 (20 QA pairs): $\lambda_{\text{final}} \approx 2.4$
- TOFU 1B fgt05 (100 QA pairs): $\lambda_{\text{final}} \approx 3.6$
- TOFU 1B fgt10 (200 QA pairs): $\lambda_{\text{final}} \approx 4.5$
- MUSE Books (554 documents): $\lambda_{\text{final}} \approx 2.3$

Table 25: Complete ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE (A0). LLM judge: FL = forget leakage, RA = retain accuracy, ret_RQ = retain response quality (all 0–2 scale).

ID	Configuration	fk↓	vm↓	ret↑	HM↑	Δ ret	FL _J ↓	RA _J ↑	rRQ _J ↑	HM _J ↑
A0	Full BLADE	.110	.000	.658	.823	—	0.14	1.40	1.69	.814
A1	$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
A9	$\tau=1.0$ (unclamped)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
A15	No LoRA (full FT)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
A16	ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
A18	Swapped bilevel	.372	.643	.651	.506	−.007	—	—	—	—
A4	GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
A5	NPO forget loss	.457	.997	.662	.009	+ .004	1.44	1.47	1.72	.495
A17	Logit margin loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

Table 26: KnowUndo full results (Llama-2-7B-chat, seed=42). HM = hmean(1−fgt_ROUGE, ret_ROUGE, MMLU). LLM judge scores on 0–1 scale (Claude Sonnet 4.6).

Method	Copyright						LLM Judge		
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	RQ↑
FT (target)	.854	.244	.894	.336	.439	.456	.602	.830	.795
GradDiff	.000	.000	.623	.200	.387	.349	.000	.344	.239
NPO	.670	.193	.757	.300	.447	.441	.243	.783	.740
SimNPO	.245	.012	.786	.318	.435	.464	.000	.755	.556
RMU	.444	.081	.768	.207	.390	.353	.000	.453	.479
PDU	.051	.007	.846	.302	.441	.456	.007	.748	.500
BLADE	.133	.009	.848	.334	.442	.479	.021	.875	.612

Method	Privacy						LLM Judge		
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	RQ↑
FT (target)	.940	.665	.949	.698	.445	.450	.882	.847	.975
GradDiff	.070	.056	.753	.427	.434	.526	.028	.495	.379
NPO	.634	.313	.786	.431	.449	.500	.282	.555	.897
SimNPO	.486	.289	.889	.540	.441	.543	.177	.657	.548
RMU	.578	.072	.669	.084	.361	.190	.009	.014	.081
PDU	.059	.032	.809	.468	.441	.552	.018	.574	.379
BLADE	.209	.075	.920	.621	.435	.601	.068	.732	.477

Method	fk↓	vm↓	rk↑	HM↑
Gold	.324	.204	.552	.660
GradDiff	.329 $\pm$.03	.043 $\pm$.02	.269 $\pm$.02	.479 $\pm$.02
NPO	.517 $\pm$.02	.274 $\pm$.02	.435 $\pm$.01	.522 $\pm$.01
SimNPO	.628 $\pm$.01	.542 $\pm$.01	.513 $\pm$.01	.440 $\pm$.00
RMU	.495 $\pm$.01	.267 $\pm$.01	.434 $\pm$.01	.531 $\pm$.00
PDU	.525 $\pm$.02	.092 $\pm$.07	.508 $\pm$.04	.577$\pm$.01
BLURNPO (seed=42)	.317	.181	.304	.448
BLADE (seed=42)	.550	.173	.475	.542

Table 27: MUSE News (5-fold, seeds=42/123/456/789/1024). BLURNPO OOM on 4×40GB.

Method	FL↓	RA↑	rRQ↑	HM _J ↑
GradDiff	0.58 $\pm$.02	0.79 $\pm$.04	1.16 $\pm$.14	.529 $\pm$.030
NPO	1.20 $\pm$.03	1.03 $\pm$.02	1.58 $\pm$.01	.531 $\pm$.011
SimNPO	1.49 $\pm$.01	1.24 $\pm$.01	1.72 $\pm$.01	.447 $\pm$.007
RMU	1.13 $\pm$.01	1.15 $\pm$.02	1.60 $\pm$.01	.565 $\pm$.004
PDU	0.87 $\pm$.12	1.18 $\pm$.05	1.64 $\pm$.05	.638$\pm$.019
BLURNPO (seed=42)	0.47	0.51	1.47	.455
BLADE (seed=42)	0.85	0.88	1.72	.579

Table 28: MUSE News LLM judge (5-fold). HM_J = hmean(1−FL/2, RA/2, rRQ/2).

sizes.

Larger forget sets require more aggressive forgetting, which causes larger retain spikes, which in turn ratchet λ higher. This automatic scaling requires no hyperparameter adjustment between settings—the same initial $\lambda_0 = 1.0$ and $\rho = 0.1$ produce appropriate constraint enforcement across a $10\times$ range of forget set

λ plateau as convergence diagnostic. Once λ reaches its plateau (visible as a flat segment in the dynamics plots), the system has found a stable balance between forgetting and retention. This provides a *built-in convergence diagnostic* without requiring held-out validation: if λ is still rising, the model has not

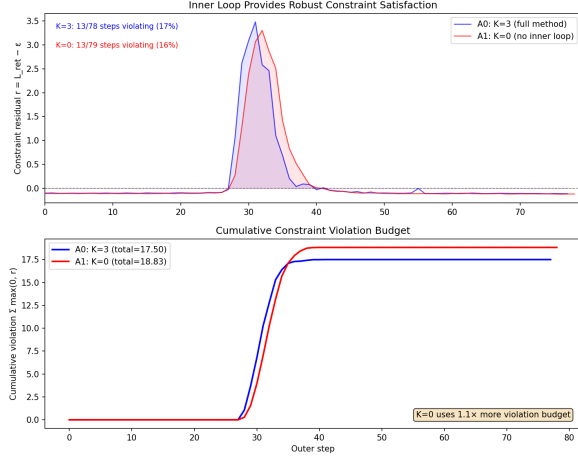


Figure 6: Constraint residual $r = L_{\text{ret}} - \epsilon$ over training. Both $K=3$ and $K=0$ have the same number of violating steps (13), but $K=0$'s violations extend later in training (wider red shading), producing slightly more cumulative damage. After step 50, both methods achieve permanent constraint satisfaction.

yet found an equilibrium; if λ is flat, training can be safely stopped. In practice, λ plateaus 10–15 steps after the spike, coinciding with $\mathcal{L}_{\text{fgr}}$ dropping below 1.0.

D.3 PDU Instability: Oscillatory Limit Cycles

PDU (Entesari et al., 2025) uses an unbounded logit-margin loss with symmetric primal-dual updates. Our dynamics analysis reveals a structural instability:

Unbounded loss $\Rightarrow$ divergent gradients. PDU's forget loss (logit margin) has no upper bound. As training progresses and the model becomes more confident on forget tokens, the margin loss grows, producing increasingly large outer perturbations. Unlike BLADE's clamped entropy (bounded by $\tau \cdot H_{\text{max}}$), there is no mechanism to attenuate the forget gradient as forgetting succeeds.

Symmetric dual $\Rightarrow$ no ratchet. PDU's dual variable uses symmetric updates: it rises during violation and falls equally fast during satisfaction. After a retain spike ratchets the dual up, subsequent satisfaction immediately erodes the protection. This creates a limit cycle: large perturbation $\rightarrow$ retain spike $\rightarrow \lambda$ rises $\rightarrow$ retain recovers $\rightarrow \lambda$ falls $\rightarrow$ large perturbation again.

Scalability (single-shot, increasing forget size)

Scale	Method	fk↓	vm↓	rk↑	HM↑
1× (889)	PDU	.542	.065	.522	.581
	BLADE	.550	.173	.475	.542
2× (1778)	PDU	.495	.050	.467	.579
	BLADE	.504	.321	.500	.547
3× (2667)	PDU	.538	.024	.488	.573
	BLADE	.466	.366	.505	.552
4× (3554)	PDU	.321	.021	.002	.005
	BLADE		(running)		

Sustainability (sequential unlearning steps)

Step	Method	fk↓	vm↓	rk↑	HM↑
1	PDU	.542	.065	.522	.580
	BLADE	.550	.173	.475	.542
2	PDU	.554	.007	.470	.558
	BLADE	.539	.212	.485	.545
3	PDU	.550	.001	.005	.016
	BLADE	.556	.230	.459	.523
4	PDU	.591	.003	.051	.130
	BLADE		(running)		

Table 29: Scalability and sustainability on MUSE News. PDU collapses at 4× scale and after step 2; BLADE maintains stable HM through step 3.

Empirical signature. On MUSE News, PDU's retain loss oscillates between 0.8 and 3.0 throughout training, never achieving the stable sub- ϵ convergence that BLADE exhibits after its single ratchet event. On MUSE Books, PDU shows 5+ distinct retain spikes across 4 epochs, each partially repaired but never fully resolved. These oscillations directly cause the worse HM scores: each spike damages shared representations, and while the token-level metrics may recover, the generation quality (measured by LLM judge) does not fully bounce back.

Implication for scalability. The oscillatory instability compounds under repeated application (sustainability) and larger forget sets (scalability). Each sequential unlearning step starts from a model that has accumulated representational damage from prior oscillations. By step 3, PDU's retain loss diverges entirely (HM=0.016). BLADE's monotonic λ ratchet prevents this accumulation: each application starts from a model where λ has permanently encoded the protection level needed.

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